

8.1 INTRODUCTION

In this chapter we deal with mapping by general analytic functions $w = f(z)$. A special class of mappings will be defined to be conformal mappings. These mappings are important in Engineering Mathematics in solving various problems in two dimensional potential theory.

8.2 MAPPING OR TRANSFORMATION

If $y = f(x)$ is a real valued function, then the equation $y = f(x)$ gives a relationship between the points on x -axis and points on y -axis. This relationship can be represented by a curve drawn in xy -plane. But when $f(z)$ is a complex valued function of a complex variable z , no such convenient graphical representation is possible, because z and w require two different planes to represent them.

However by representing $z = x + iy$ in z -plane and the corresponding $w = u + iv$ in w -plane, we can establish a graphical correspondence between the two planes.

8.3 DEFINITION

The correspondence defined by the equation $w = f(z)$ or $u = u(x, y)$ and $v = v(x, y)$ between the points in the z -plane and w -plane is called a "mapping" or transformation from z -plane to the w -plane.

The corresponding points w in the w -plane are called the "images" of the points z of the z -plane.

If for every point of z -plane, there corresponds one and only one point of w -plane, we say that the correspondence is one-to-one.

8.4 DEFINITION. CONFORMAL MAPPING

[JNTU 2005S, Nov. 2006, JNTU (A), (H) Nov. 2009 (Set No. 1,2,(3))]

Suppose under the transformation $w = f(z)$, i.e., $u = u(x, y)$ and $v = v(x, y)$, the point $P(x_0, y_0)$ of the z -plane is mapped into the point $P'(u_0, v_0)$ of the w -plane. Suppose c_1 and c_2 are any two curves intersecting at the point $P(x_0, y_0)$. Suppose the mapping takes c_1 and c_2 into the curves c'_1 and c'_2 which are intersecting at $P'(u_0, v_0)$. If the transformation is such that the angles between c_1 and c_2 at (x_0, y_0) is equal both in magnitude and sense to the angle between c'_1 and c'_2 at (u_0, v_0) , then it is said to be a conformal transformation at (x_0, y_0) . (See figure (8.1) and (8.2)).

If the transformation preserves the magnitudes but not necessarily sense, then it is called bi-conformal.

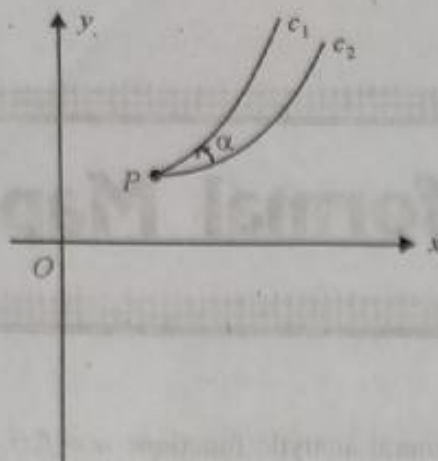


Fig. 8.1.

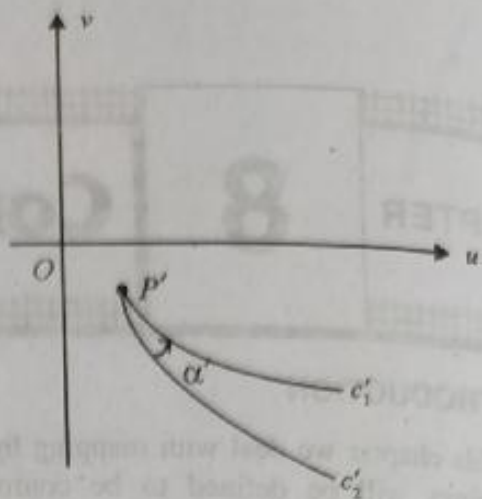


Fig. 8.2.

8.5 SUFFICIENT CONDITIONS FOR $w = f(z)$ TO REPRESENT A CONFORMAL MAPPING

Theorem : Let $f(z)$ be an analytic function of z in a domain D of the z -plane and let $f'(z) \neq 0$ in D . Then $w = f(z)$ is a conformal mapping at all points of D .

[JNTU 2005S, Nov. 2006, Nov.2008S, JNTU (A), (H) Nov. 2009 (Set No.1,2,3)]

Proof : Let $P(z)$ be a point in the region R of the z -plane and $P'(w)$ is the corresponding point in the region R' of the w -plane. Suppose z moves on a curve c and w moves on the corresponding curve c' . Let $Q(z + \Delta z)$ be a neighbouring point on c and $Q'(w + \Delta w)$ be the corresponding point on c' . Suppose $\overline{PQ} = \Delta z$ and $\overline{P'Q'} = \Delta w$. Then $\Delta z = re^{i\theta}$ where $|\Delta z| = r = \text{length of } PQ$.

$\theta = \text{amplitude} = \text{angle made by } PQ \text{ with } x\text{-axis.}$

Similarly, let $\Delta w = r'e^{i\theta'}$ where $r' = |\Delta w|$, $\theta' = \text{amplitude}$.

We can have
$$\frac{\Delta w}{\Delta z} = \frac{r'}{r} e^{i(\theta' - \theta)}$$

Suppose the tangents to the curves c and c' at P and P' make angles α and α' with x -axis and u -axis respectively.

As $\Delta z \rightarrow 0$, $\theta \rightarrow \alpha$ and $\theta' \rightarrow \alpha'$

$$\therefore f'(z) = \frac{dw}{dz} = \lim_{\Delta z \rightarrow 0} \left(\frac{r'}{r} \right) e^{i(\alpha' - \alpha)} \quad \dots(1)$$

Let β and ϕ be the modulus and amplitude of the function $f'(z) (\neq 0)$.

$$\text{Then } f'(z) = \beta e^{i\phi} \quad \dots(2)$$

$$\text{From (1) and (2), we obtain } \beta = \lim_{\Delta z \rightarrow 0} \frac{r'}{r} \quad \dots(3)$$

$$\text{and } \phi = \alpha' - \alpha \quad \dots(4)$$

Let c_1 be another curve through β in the z -plane and c'_1 be the corresponding curve through P' in the w -plane.

Let the tangents at P and P' to the curves c_1 and c'_1 make angles β and β' with x -axis and u -axis.

Then, we have $\phi = \beta' - \beta$ as in (4).

From (4) and (5), we get

$$\alpha' - \alpha = \beta' - \beta \Rightarrow \beta - \alpha = \beta' - \alpha' = \nu$$

...(5)

Thus, we can see the angle between curves (say) and their images c' and c'_1 is preserved in magnitude and direction through the c and c_1 transformation.

Hence the theorem.

Note. If $f'(z) = 0$ at a point, then that point is called a "critical point" of the function.

e.g. (1). Consider $f(z) = z^2$. It has a critical point at $z = 0$.

e.g. (2). Similarly, $f(z) = \cos z \Rightarrow f'(z) = -\sin z$

If $f'(z) = 0$ then $\sin z = 0$ and critical points of $\cos z$ are $n\pi$, $n \in \mathbb{Z}$.

8.6 STANDARD TRANSFORMATIONS

1. Translation. $w = z + c$ where c is any complex constant is a translation.

Suppose $z = x + iy$, $c = c_1 + ic_2$ and $w = u + iv$.

Then, we get $u + iv = (x + iy) + (c_1 + ic_2)$.

Comparing real and imaginary parts,

$$u = x + c_1 \quad \text{and} \quad v = y + c_2$$

i.e., if $P(x, y)$ is a point in the z -plane then $P'(x + c_1, y + c_2)$ is the corresponding point in the w -plane. Thus, if the w -plane is superimposed on z -plane, the figure is shifted through a distance which is given by c . Thus, this transformation maps a figure in the z -plane to a figure of same shape and size in w -plane. In particular circles are mapped onto circles under this transformation.

2. Expansion or contraction and rotation: (Magnification)

Consider the transformation $w = cz$, where c is the complex constant.

Let $z = re^{i\theta}$, $w = Re^{i\phi}$ and $c = \beta e^{i\alpha}$. Then

$$w = cz \Rightarrow Re^{i\phi} = \beta r e^{i(\theta + \alpha)}$$

We get $R = \beta r$ and $\phi = \theta + \alpha$.

Thus, under this transformation a point $P(r, \theta)$ in the z -plane is mapped to the point $P'(\beta r, \theta + \alpha)$ in the w -plane. Thus, this transformation effects an expansion when $|c| > 1$ and a contraction when $0 < |c| < 1$ of the radius vector by $\beta = |c|$ and rotation through an angle $\alpha = \text{amp } c$. Hence, any figure in z -plane is transformed into, geometrically, a similar figure in the w -plane. In particular circles are mapped to circles.

Note (1). The above two transformations are the special cases of the transformation

$w = az + c$, where a and c are complex constants. Put $c = 0$, we get $w = az$, and by putting $a = 1$, we get $w = z + c$. This transformation $w = az + c$ is called a linear transformation.

Note (2). When $|c| = 1$ then $w = cz$ is called a pure rotation, since in this case there is no expansion or contraction, but just a rotation through an angle of α .

Result : Circles are invariant under linear transformation.

Sol. Consider the linear transformation $w = az + c$ where a and c are complex constants.

Consider the circle

$$A(x^2 + y^2) + Bx + Cy + D = 0 \quad \dots (1)$$

We have $u + iv = w = az + c = a(x + iy) + C_1 + iC_2$

Comparing the real and imaginary parts,

$$u = ax + C_1$$

$$\text{and } v = ay + C_2$$

$$\Rightarrow x = \frac{u - C_1}{a}, y = \frac{v - C_2}{a} \quad \dots (2)$$

Substituting in (1), we get

$$A'(u^2 + v^2) + B'u + C'v + D' = 0 \quad \dots (3)$$

which is a circle in the w -plane where

$$A' = \frac{A}{a^2}, B' = \frac{B - 2AC_1}{a}, C' = \frac{C - 2AC_2}{a}, D' = D + A\left(\frac{C_1^2 + C_2^2}{a^2}\right) - \frac{BC_1}{a} - \frac{CC_2}{a}$$

Hence the result.

3. The mapping $w = \frac{1}{z}$ (is called inversion) :

Suppose $z = re^{i\theta}$ and $w = Re^{i\phi}$. Then $w = \frac{1}{z}$ gives $Re^{i\phi} = \frac{1}{re^{i\theta}}$ which gives $R = \frac{1}{r}$ and $\phi = -\theta$.

From this, we can see that the image $w = \frac{1}{z}$ of a $z (\neq 0)$ lies on the ray from the origin through

$\bar{z}$, at a distance of $\frac{1}{|z|}$. Particularly, if $z = e^{i\theta}$ on the unit circle $|z| = 1$ then it is mapped onto $w = e^{-i\theta} = \bar{z}$ on the unit circle $|w| = 1$.

The transformation $w = \frac{1}{z}$ can be obtained

geometrically from z first by an inversion in the unit circle and then followed by a reflection w.r.t. x -axis.

$$OP \cdot OQ = 1$$

P' is the image of Q with respect to x -axis.

(see fig. 8.3)

$$\therefore w = \frac{1}{z} = \frac{1}{x + iy} = \frac{x - iy}{x^2 + y^2} = \frac{\bar{z}}{z\bar{z}}$$

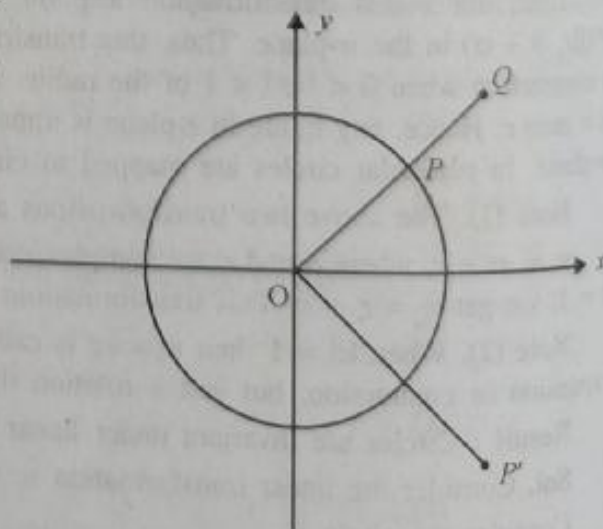


Fig. 8.3

Result. The transformation $w = \frac{1}{z}$ maps every straight line or circle onto a circle or straight line.

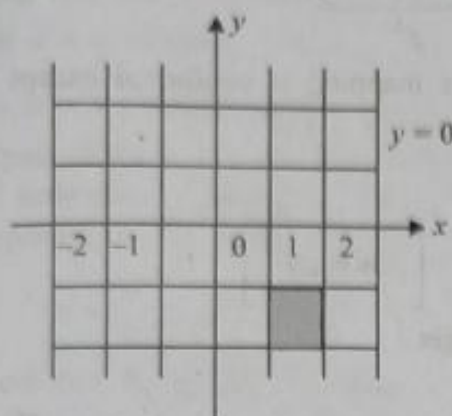


Fig. 8.4

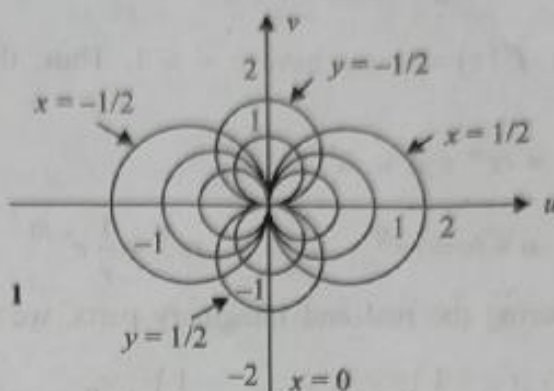


Fig. 8.5

Proof. We have that every circle or straight line in z -plane can be written as $A(x^2 + y^2) + Bx + Cy + D = 0$... (1) when A, B, C, D are real. If $A = 0$ and at least one of B and C is not 0, it becomes a straight line. If $A \neq 0$, it becomes a circle,

We have $z = x + iy$ and $\bar{z} = x - iy$ and $x^2 + y^2 = z\bar{z}$, $x = \frac{z + \bar{z}}{2}$ and $y = \frac{z - \bar{z}}{2i}$.

Thus, (1) can be written as $Az\bar{z} + B\frac{z + \bar{z}}{2} + C\frac{z - \bar{z}}{2i} + D = 0$.

Substituting $w = \frac{1}{z}$ and multiplying with $w\bar{w}$, we get

$$A \frac{1}{w\bar{w}} + B \left(\frac{\frac{1}{w} + \frac{1}{\bar{w}}}{2} \right) + C \left(\frac{\frac{1}{w} - \frac{1}{\bar{w}}}{2i} \right) + D = 0$$

$$\Rightarrow A + B \left(\frac{\bar{w} + w}{2} \right) + C \left(\frac{\bar{w} - w}{2i} \right) + D w\bar{w} = 0$$

Rearranging, we get $A + Bu - Cv + D(u^2 + v^2) = 0$.

This in general represents a straight line if $D = 0$ and a circle if $D \neq 0$ in w -plane. Thus, the general equation of a circle or straight line is transformed to the general equation of a circle or straight line.

Hence the result. Consider the bilinear transformation

8.7 SOME SPECIAL CONFORMAL TRANSFORMATIONS

1. The transformation $w = z + \frac{1}{z}$ (JOUKOWSKI'S transformation)

[JNTU 2000, 2007S, Nov. 2008 (Set No.1)]

(OR) Show that the transformation $w = z + \frac{1}{z}$ converts the radial lines $\theta = \text{constant}$ in the z -plane into a family of confocal hyperbolas in the w -plane.

[JNTU (A) Dec 2009 (Set No. 3)]

Proof : Let $w = f(z) = z + \frac{1}{z}$

Differentiating, we get $f'(z) = 1 - \frac{1}{z^2} = \frac{(z+1)(z-1)}{z^2}$

When $f'(z) = 0$, we have $z = \pm 1$. Thus, the mapping is conformal except at the points $z = \pm 1$.

Let $z = re^{i\theta}$ and $w = u + iv$.

Then, $u + iv = re^{i\theta} + \frac{1}{re^{i\theta}} = re^{i\theta} + \frac{1}{r}e^{-i\theta} \quad \left[\because w = z + \frac{1}{z} \right]$

Comparing the real and imaginary parts, we get

$$u = \left(r + \frac{1}{r}\right) \cos \theta, \quad v = \left(r - \frac{1}{r}\right) \sin \theta \quad \dots(1)$$

From which, we get $\cos \theta = \frac{u}{r + \frac{1}{r}}, \quad \sin \theta = \frac{v}{r - \frac{1}{r}}$

Substituting in $\cos^2 \theta + \sin^2 \theta = 1$, we get $\frac{u^2}{\left(r + \frac{1}{r}\right)^2} + \frac{v^2}{\left(r - \frac{1}{r}\right)^2} = 1 \quad \dots(2)$

which represents an ellipse.

Thus, circles of the form $r = \text{constant}$ in the z -plane are mapped into a family of ellipses in the w -plane.

Since $a^2 - b^2 = \left(r + \frac{1}{r}\right)^2 - \left(r - \frac{1}{r}\right)^2 = 4$ (constant)

we have the ellipses are confocal (Fig. 8.6).

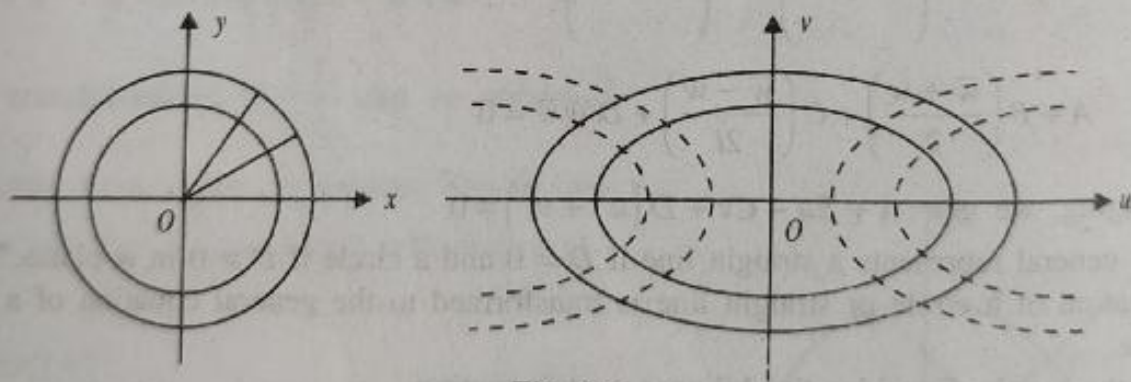


Fig. 8.6

In particular unit circle $r = 1$ gives $u = 2 \cos \theta$ and $v = 0$.

Since $-1 \leq \cos \theta \leq 1$, we have $-2 \leq u \leq 2$.

Thus, the unit circle $r = 1$ is mapped to the segment $-2 \leq u \leq 2$ of the real axis in the w -plane.

Also from (1), we have $\frac{u^2}{\cos^2 \theta} - \frac{v^2}{\sin^2 \theta} = \left(r + \frac{1}{r}\right)^2 - \left(r - \frac{1}{r}\right)^2 = 4$.

Thus, $\frac{u^2}{4 \cos^2 \theta} - \frac{v^2}{4 \sin^2 \theta} = 1$. From this we conclude that the radial lines $\theta = \text{constant}$ in

the z -plane are transformed into a family of confocal hyperbolas in the w -plane.

2. Transformation $w = z^2$

We have $w = z^2 \Rightarrow u + iv = (x + iy)^2 = x^2 - y^2 + 2ixy$.

Comparing real and imaginary parts, we have $u = x^2 - y^2$ and $v = 2xy$.

(i) If $u = a = \text{constant}$, then $x^2 - y^2 = a$ which is a rectangular hyperbola.

(ii) If $v = b = \text{constant}$, then $2xy = b \Rightarrow xy = \frac{b}{2}$ which is another rectangular hyperbola.

Thus, a pair of lines $u = a$ and $v = b$ parallel to coordinate axes in w -plane are mapped into a pair of orthogonal rectangular hyperbolas in the z -plane.

(iii) Suppose $x = c$ is a constant, then we have

$$y = \frac{v}{2c} \text{ and } y^2 = c^2 - u$$

From these two equations, we obtain

$$v^2 = 4c^2 (c^2 - u) \text{ which is a parabola.}$$

(iv) Suppose $y = d$ is a constant, then we have

$$x = \frac{v}{2d} \text{ and } u = x^2 - d^2$$

From these two equations $v^2 = 4d^2 (d^2 + u)$ which is also a parabola.

$\therefore$ The pair of lines $x = c$ and $y = d$ parallel to coordinate axes in z -plane map into orthogonal parabolas in the w -plane.

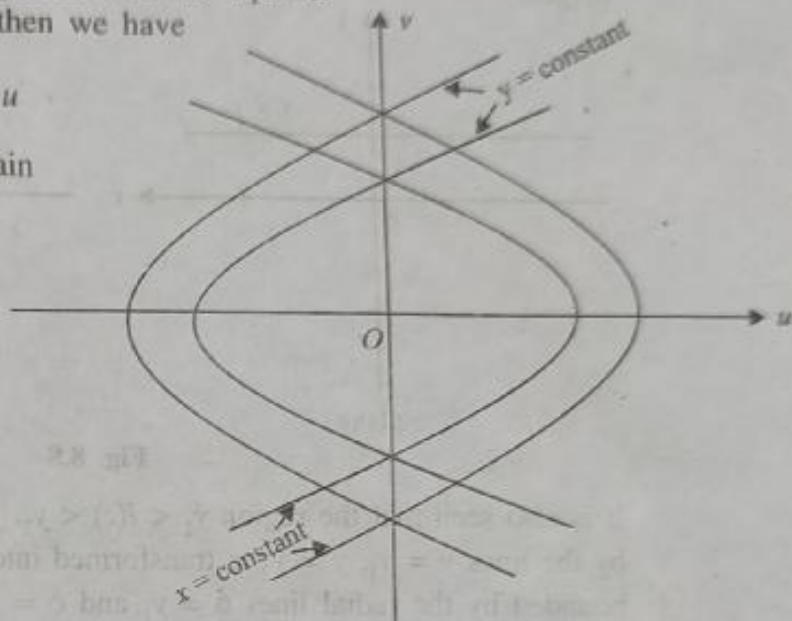


Fig. 8.7

Also $w = z^2 \Rightarrow \frac{dw}{dz} = 2z = 0 \Rightarrow z = 0 \therefore z = 0$ is a critical point.

(v) Suppose $z = re^{i\theta}$ and $w = Re^{i\phi}$ then, $w = z^2$ becomes $Re^{i\phi} = r^2 e^{2i\theta}$.

Thus, $R = r^2$ and $\phi = 2\theta$.

These points in the z -plane at (r, θ) are rotated through angle 2θ . Since all the points in the first quadrant of the z -plane occupy the region $0 \leq \theta \leq \frac{\pi}{2}$, they map into $0 \leq \phi \leq \pi$ or upper half of the w -plane. The upper half of z -plane $0 < \theta < \pi$ transforms into the entire w -plane $0 \leq \phi \leq 2\pi$. The same is true for the lower half.

3. Transformation $w = z^n$, where n is a positive integer

We have $\frac{dw}{dz} = 0 \Rightarrow nz^{n-1} = 0 \Rightarrow z = 0$.

As above $w = z^n$, $Re^{i\phi} = (re^{i\theta})^n = r^n e^{in\theta}$

$\therefore R = r^n$ and $\phi = n\theta$.

Thus, when $0 < \theta < \frac{\pi}{n}$ we have $0 < \phi < \pi$.

Thus, $w = z^n$ gives a conformal mapping of the z -plane everywhere except at the origin and it maps a sector of z -plane of central angle $\frac{\pi}{n}$ to cover the upper half of the w -plane.

4. The transformation $w = e^z$

[JNTU 2004 (Set No. 1)]

Let $z = x + iy$ and $w = \beta e^{i\phi}$. Then $w = e^z$ gives $\beta e^{i\phi} = e^{x+iy} = e^x \cdot e^{iy}$

Then $\beta = e^x \dots (1), \quad \phi = y \dots (2)$

- (i) Consider the image of the line $y = y_1$ ($0 < y_1 < 2\pi$) in the z -plane. This is a line parallel to x -axis.

As x takes values from $-\infty$ to ∞ , we see that e^x increases from 0 to ∞ monotonically. Therefore, we can see that the image of the line $y = y_1$ is the ray $\phi = y_1$ excluding the origin.

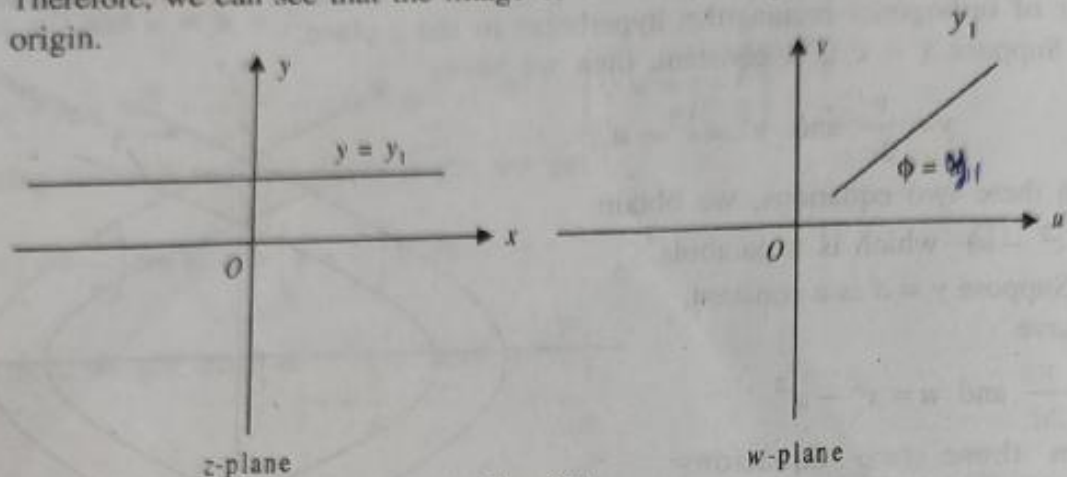


Fig. 8.8

It is also seen that the region $y_1 < I(z) < y_2$, $y_2 - y_1 < \pi$, i.e., the infinite strip bounded by the lines $y = y_1$, $y = y_2$ is transformed into the wedge shaped region in the w -plane bounded by the radial lines $\phi = y_1$ and $\phi = y_2$.

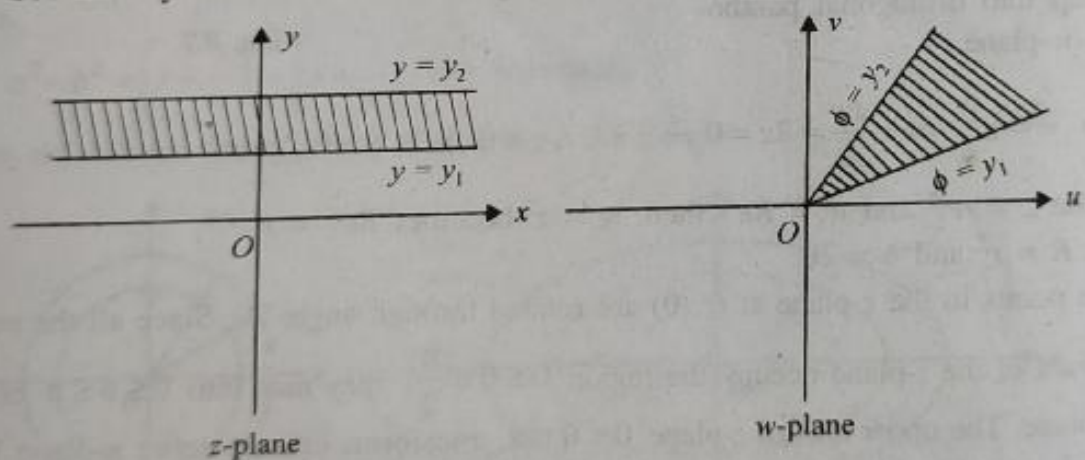


Fig. 8.9

In particular the infinite strip bounded by the lines $y = 0$ and $y = \pi$ in the z -plane is mapped into the upper half of the w -plane. [JNTU (H) Nov. 2009 (Set No. 2)]

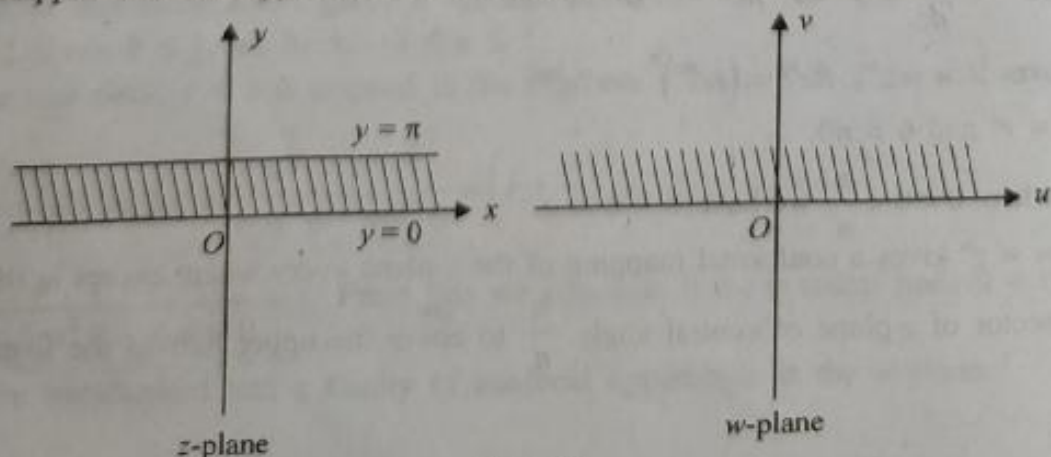


Fig. 8.10

(ii) Suppose we consider the straight line $x = x_1$.

Then from $\beta = e^x$, we get $\beta^2 = e^{2x}$

$$\Rightarrow u^2 + v^2 = e^{2x} \text{ which is a circle.}$$

Thus, any line parallel to y -axis in the z -plane is mapped into circle in the w -plane.

The infinite strip bounded by the lines $x = x_1$, $x = x_2$, is transformed into region between concentric circles.

(iii) From (i) and (ii) discussed above, we get that the rectangular region $x_1 \leq x \leq x_2$, $y_1 \leq y \leq y_2$ is mapped onto the region $e^{x_1} \leq \beta \leq e^{x_2}$, $y_1 \leq \phi \leq y_2$ bounded by circles and rays.

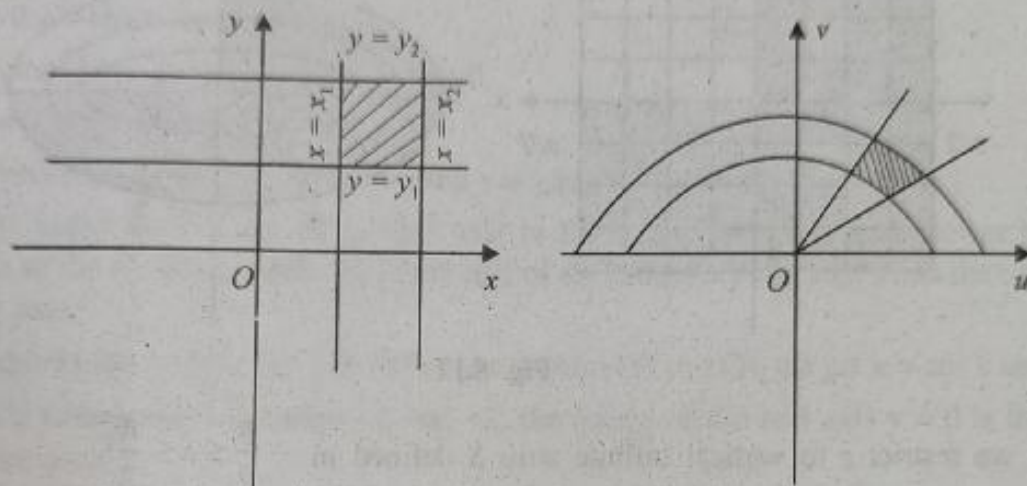


Fig. 8.11

(iv) Writing $w = \beta e^{i\phi}$ and $z = r e^{i\theta}$, the transformation i.e. $w = e^z$ becomes

$$\beta e^{i\phi} = e^{re^{i\theta}} = e^{r(\cos \theta + i \sin \theta)} = e^{r \cos \theta} \cdot e^{ir \sin \theta}$$

Thus, we get $\beta = e^{r \cos \theta}$ and $\phi = r \sin \theta$.

Eliminating r , we get $\beta = e^{\phi \cot \theta}$.

Hence, the straight lines in the z -plane defined by $\theta = \alpha$ are mapped to logarithmic spirals $\beta = e^{\phi \cot \alpha}$ for varying values of α .

5. The logarithmic transformation $w = \log z$.

Writing $z = x + iy$ and $w = u + iv$, the transformation $w = \log z$ becomes

$$u + iv = \log(x + iy)$$

$$\Rightarrow e^{u+iv} = x + iy \Rightarrow e^u \cdot e^{iv} = x + iy$$

$$\Rightarrow e^u (\cos v + i \sin v) = x + iy$$

Thus, $x = e^u \cos v$ and $y = e^u \sin v$.

Hence, the straight lines $x = x_1$, and $y = y_1$ in the z -plane correspond respectively to the curves in w -plane defined by

$$x_1 = e^u \cos v \text{ and } y_1 = e^u \sin v.$$

From the given transformation, we get $\frac{dw}{dz} = \frac{1}{z}$. At origin $z = 0$, the derivative is infinite.

Therefore, the mapping is conformal except at $z = 0$ (origin).

6. Conformal mapping $w = \sin z$ Suppose $w = \sin z$

$$\Rightarrow u + iv = \sin x \cosh y + i \cos x \sinh y \quad \dots(1)$$

$$\text{Thus, } \begin{cases} u = \sin x \cosh y \\ v = \cos x \sinh y \end{cases} \quad \dots(2)$$

We know that $\sin z$ is periodic and hence it cannot be a 1-1 function if considered in the entire z -plane.

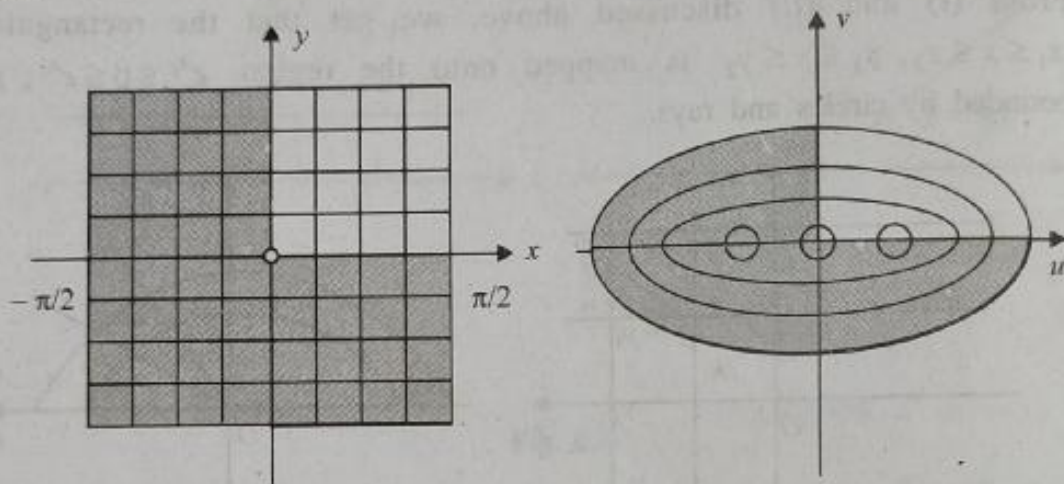


Fig. 8.12

Thus, we restrict z to vertical infinite strip S defined in $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.

Also $f'(z) = \cos z$, $f'(z) = 0 \Rightarrow z = \pm \frac{\pi}{2}$ and these are two critical points for the conformal mapping. We will consider the images of the vertical line $x = \text{constant}$. When $x = 0$, then $u = 0$ and $v = \sinh y$. Therefore, y -axis ($x = 0$) is mapped onto v -axis ($u = 0$).

When $x = \pm \frac{\pi}{2}$ then $u = \pm \cosh y$ and $v = 0$.

Since $\cosh y \geq 1$ the vertical boundaries $x = \pm \frac{\pi}{2}$ of the strip S are mapped onto the portions $u \leq -1$ and $u \geq 1$ of the u -axis.

$$\text{If } x \neq 0, \pm \frac{\pi}{2} \text{ then from (2), we get } \frac{u^2}{\sin^2 x} - \frac{v^2}{\cos^2 x} = 1. \quad \dots(3)$$

When $x \neq 0$, these are hyperbolas.

We will now consider the horizontal line $y = \text{constant}$. [JNTU Nov. 2008 (Set No. 3)]

If $y = 0$, then, $\sinh y = 0$ hence, $u = \sin x$ and $v = 0$.

Thus, the x -axis is mapped onto the segment $-1 \leq u \leq 1$ of the u -axis.

$$\text{If } y \neq 0, \text{ from (2), we get } \frac{u^2}{\cosh^2 y} + \frac{v^2}{\sinh^2 y} = 1. \quad [\text{Since } \sin^2 x + \cos^2 x = 1]$$

These are conformal ellipses with foci ± 1 .

The images of curves of $x = \text{constant}$ and $y = \text{constant}$ form an orthogonal net, except at the critical points $z = \pm \frac{\pi}{2}$.

Thus, from the above discussion $w = \sin z$ maps the infinite vertical strip $-\frac{\pi}{2} < x < \frac{\pi}{2}$ onto the w -plane cut along the rays $u \leq -1$ and $u \geq 1, v = 0$. This mapping is 1-1 and conformal.

[JNTU 2005S, June 2009 S (Set No. 3)]

7. Conformal mapping $w = \cos z$

Sol. We have

$$w = u + iv = \cos(x + iy) = \cos x \cos iy - \sin x \sin iy = \cos x \cosh y - i \sin x \sinh y$$

$$\text{Hence } u = \cos x \cosh y$$

$$\text{and } v = -\sin x \sinh y$$

... (1)

The imaginary axis in the z -plane is $x = 0$.

... (2)

When $x = 0$, $u = \cosh y$ and $v = 0$

If $y > 0$, $\sinh y$ is positive and if $y < 0$, $\sinh y$ negative.

Hence the upper half of the imaginary axis in the z -plane maps into the lower half of the imaginary axis of the w -plane, while the lower half of the imaginary axis maps into the upper half of the imaginary axis.

The real axis in the z -plane is $y = 0$. When $y = 0$, from (1) and (2), we get $u = \cos x$ and $v = 0$. Since $\cos x$ takes values between -1 and $+1$, the image of the real axis $y = 0$ is the segment $-1 \leq u \leq 1$ of the u -axis.

$$\text{Eliminating } x \text{ from (1) and (2), we get } \frac{u^2}{\cosh^2 y} + \frac{v^2}{\sinh^2 y} = 1$$

$$\text{When } y = a \text{ constant} = K(\text{say}), \text{ we get } \frac{u^2}{\cosh^2 K} + \frac{v^2}{\sinh^2 K} = 1 \quad \dots (3)$$

When $K \neq 0$, (3) represents a system of ellipses having foci at $(1, 0)$ and $(-1, 0)$ with semi-axes $\cosh(K)$ and $\sinh(K)$.

Thus, lines parallel to the real axis of the z -plane maps into confocal ellipses in the w -plane.

Eliminating y from (1) and (2), we get

$$\frac{u^2}{\cos^2 x} - \frac{v^2}{\sin^2 x} = 1$$

When $x = a \text{ constant} = c$, we get

$$\frac{u^2}{\cos^2 c} - \frac{v^2}{\sin^2 c} = 1 \quad \dots (4)$$

When $c \neq 0$, (4) represents a system of hyperbolas having foci at $(1, 0)$ and $(-1, 0)$. Hence lines parallel to the imaginary axis of the z -plane maps into confocal hyperbolas.

Note. The mapping $w = \cos z$ can be discussed independently but $w = \cos z = \sin\left(\frac{\pi}{2} + z\right)$. This is same as the mapping $w = \sin z$ preceded by a translation to the right through $\frac{\pi}{2}$ units in the z -plane.

8. The conformal mapping $w = \tan z$

Taking $z = x + iy$ and $w = u + iv$, we obtain

$$\begin{aligned} u + iv = \tan(x + iy) &= \frac{\sin(x + iy)}{\cos(x + iy)} = \frac{2 \sin(x + iy) \cos(x - iy)}{2 \cos(x + iy) \cos(x - iy)} \\ &= \frac{\sin 2x + i \sinh 2y}{\cos 2x + \cosh 2y} \end{aligned}$$

Equating the real and imaginary parts, we obtain

$$u = \frac{\sin 2x}{\cos 2x + \cosh 2y} \quad \text{and} \quad v = \frac{\sinh 2y}{\cos 2x + \cosh 2y}$$

Consider the line $x = x_1$, we get

$$u = \frac{\sin 2x_1}{\cos 2x_1 + \cosh 2y} \quad \text{and} \quad v = \frac{\sinh 2y}{\cos 2x_1 + \cosh 2y}$$

with y as parameter.

This can be written as

$$u \cosh 2y = \sin 2x_1 - u \cos 2x_1 \quad \dots(1)$$

$$\text{and} \quad \sinh 2y = v \cos 2x_1 + v \cosh 2y$$

$$= v \cos 2x_1 + v \left(\frac{\sin 2x_1}{u} - \cos 2x_1 \right) \quad [\text{by (1)}]$$

$$\text{or} \quad u \sinh 2y = v \sin 2x_1 \quad \dots(2)$$

We eliminate y from (1) and (2). By squaring and subtracting equations (1) and (2), we get

$$u^2 (\cosh^2 2y - \sinh^2 2y) = \sin^2 2x_1 + u^2 \cos^2 2x_1 - 2u \sin 2x_1 \cos 2x_1 - v^2 \sin^2 2x_1$$

$$\text{or} \quad u^2 (\sin^2 2x_1) + v^2 \sin^2 2x_1 + 2u \sin 2x_1 \cos 2x_1 - \sin^2 2x_1 = 0$$

$$\Rightarrow \quad u^2 + v^2 + 2u \cot 2x_1 - 1 = 0.$$

This represents a circle for different values of x_1 , whose centre is $(-\cot 2x_1, 0)$ and radius is $\sqrt{\cot^2 2x_1 + 1} = \operatorname{cosec} 2x_1$.

All these circles pass through the points $(0, \pm 1)$, i.e., through the point $w = \pm i$.

The infinite strip $x_1 < x < x_2$ ($0 < x_2 - x_1 \leq \pi$) parallel to the real axis in the z -plane is mapped into a circular line with angles $2(x_2 - x_1)$ radians and vertices at $w = \pm 1$.

Similarly, we can see that the straight lines $y = y_1$ for varying values of y_1 is mapped onto the family of circles $u^2 + v^2 - 2v \cot \alpha 2y_1 + 1 = 0$. (See Fig. 8.13)

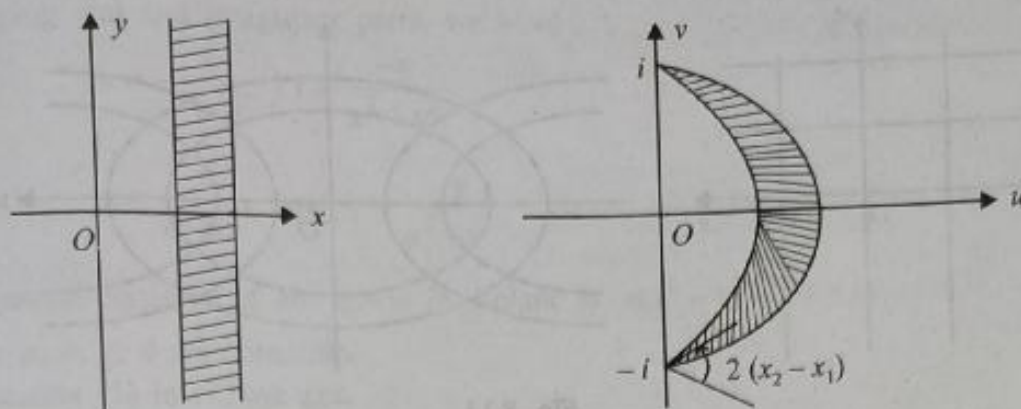


Fig. 8.13

9. Transformation $w = \sinh z$

The transformation is

$$w = \sinh z = \frac{1}{i} \sin iz \quad [\because \sin iz = i \sinh z]$$

or $iw = \sin(iz)$

Letting $W = iw$ and $Z = iz$, we have
 $W = \sin Z$.

The transformation $Z = iz$ is a rotation of every point in the z -plane through an angle $\frac{\pi}{2}$ in the anti-clockwise direction. Similarly the transformation $W = iw$ is a rotation of every point in the w -plane through an angle $\frac{\pi}{2}$ in the anti-clockwise direction. We have discussed in detail the transformation $w = \sin z$.

Hence the transformation $w = \sinh z$ is the same as the transformation $w = \sin z$ with a rotation of the axes in the z and w -planes through an angle $\frac{\pi}{2}$ in the counter-clockwise direction.

[JNTU 2003S (Set No. 4)]

10. Transformation $w = \cosh z$

Let $w = f(z) = \cosh z$. Then,

$$\begin{aligned} u + iv &= \cosh(x + iy) = \cosh x \cosh iy + \sinh x \sinh iy \\ &= \cosh x \cos y + i \sinh x \sinh y \end{aligned}$$

Comparing, we get

$$u = \cosh x \cos y \quad \dots(1)$$

and

$$v = \sinh x \sin y \quad \dots(2)$$

Eliminating x ,

$$\frac{u^2}{\cosh^2 y} - \frac{v^2}{\sinh^2 y} = 1 \quad \dots(3)$$

and eliminating y ,

$$\frac{u^2}{\cosh^2 x} + \frac{v^2}{\sinh^2 x} = 1 \quad \dots(4)$$

From (3), lines parallel to x -axis in the z -plane are mapped into hyperbolas in the w -plane.

From (4), straight lines parallel to y -axis in the z -plane are mapped into ellipses in the w -plane.

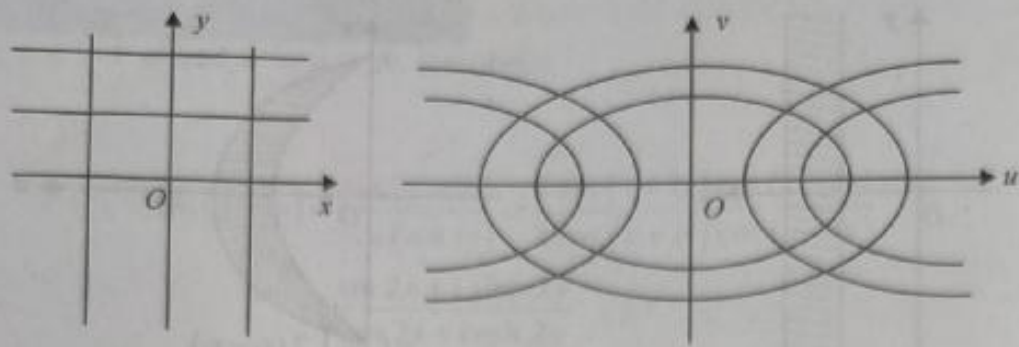


Fig. 8.14

Note :

1. We have $w = \sinh z = -i \sin(iz)$

If we take $w_1 = iz$, $w_2 = \sin(w_1)$. Then $w = -iw_2$

Thus the transformation $\sinh z$ consists of anti-clockwise rotation through $\pi/2$ followed by $\sin$ transformation

2. $w = \cosh z = \cos(iz)$

Taking $w_1 = iz$ then $w = \cos(w_1)$

Thus $\cosh z$ transformation consists of anti-clockwise rotation through $\pi/2$ followed by $\cos$ transformation.

SOLVED EXAMPLES

Example 1 : Under the transformation $w = \frac{1}{z}$ find the image of the circle $|z - 2i| = 2$.

[JNTU 2001, 2004, April 2006, Aug. 2007S, JNTU (K), (H) Nov. 2009 (Set No. 1

Solution : Given $u + iv = w = \frac{1}{z} = \frac{1}{x + iy} = \frac{x - iy}{x^2 + y^2}$

$$\therefore x = \frac{u}{u^2 + v^2}, y = \frac{-v}{u^2 + v^2}$$

$|z - 2i| = 2$ is a circle in z -plane passing through origin.

$$\therefore |x + iy - 2i|^2 = x^2 + (y - 2)^2 = 2^2 = 4$$

$$\Rightarrow x^2 + y^2 - 4y + 4 = 4$$

Substituting for x and y , we get

$$\frac{u^2}{(u^2 + v^2)^2} + \frac{v^2}{(u^2 + v^2)^2} + \frac{4v}{(u^2 + v^2)} = 0$$

$$\Rightarrow 1 + 4v = 0 \text{ i.e. } v = -\frac{1}{4}$$

which is a straight line in the w -plane.

Example 2 : Show that the transformation $w = \frac{1}{z}$ maps a circle to a circle or to a straight line if the former goes through the origin. [JNTU 2000]

Solution : $u + iv = \frac{1}{z} = w = \frac{1}{x + iy} = \frac{x - iy}{x^2 + y^2}$

Comparing real and imaginary parts, we have

$$u = \frac{x}{x^2 + y^2}, \quad v = \frac{-y}{x^2 + y^2}$$

$$\Rightarrow x = \frac{u}{u^2 + v^2} \quad \text{and} \quad y = \frac{-v}{u^2 + v^2} \quad \dots (1)$$

$$\text{The general equation of any circle in } z\text{-plane is } a(x^2 + y^2) + bx + cy + d = 0 \quad \dots (2)$$

Where a, b, c, d are constants.

Substituting (1) in (2) we get,

$$a \left[\left(\frac{u}{u^2 + v^2} \right)^2 + \left(\frac{-v}{u^2 + v^2} \right)^2 \right] + \frac{bu}{u^2 + v^2} - \frac{cv}{u^2 + v^2} + d = 0$$

$$\Rightarrow d(u^2 + v^2) + bu - cv + a = 0 \quad \dots (3)$$

Equation (3) represents a circle in w -plane. Thus a circle (1) in z -plane transforms to a circle (3) in w -plane. If the circle (1) in z -plane passes through origin then $d = 0$, so the image is $bu - cv + a = 0$ is a straight line in w -plane.

Example 3 : Find the image of the infinite strip $0 < y < \frac{1}{2}$ under the transformation $w = \frac{1}{z}$.

[JNTU A 2005, 2007S, Aug. 2007S, 2008S, Nov. 2008, JNTU (K) Nov. 2009 (Set No. 1)]

Solution : Given $w = \frac{1}{z} \Rightarrow z = \frac{1}{w}$

$$\Rightarrow x + iy = \frac{1}{u + iv} = \frac{u - iv}{u^2 + v^2}$$

Comparing real and parts imaginary, we get, $x = \frac{u}{u^2 + v^2}, y = \frac{-v}{u^2 + v^2}$

Suppose $y = 0 \Rightarrow v = 0$

Suppose $y = \frac{1}{2}$ then $\frac{1}{2} = \frac{-v}{u^2 + v^2} \Rightarrow u^2 + v^2 + 2v = 0$

$\Rightarrow u^2 + (v + 1)^2 = 1$ which is a circle with centre $(0, -1)$ and radius 1.

Hence under transformation $w = \frac{1}{z}$ the straight line $y = 0$ is transformed to line $v = 0$ and the line

$y = \frac{1}{2}$ is transformed to circle $u^2 + (v + 1)^2 = 1$

Hence the infinite strip $0 < y < \frac{1}{2}$ in z -plane is mapped into the region between the line $v = 0$ and

the circle $u^2 + (v + 1)^2 = 1$ is w -plane under the transformation $w = \frac{1}{z}$.

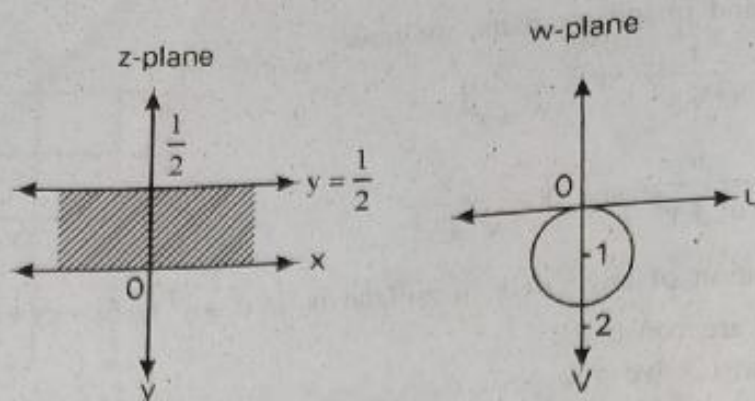


Fig. 8.15

Example 4 : Show that the image of the hyperbola $x^2 - y^2 = 1$ under the transformation $w = \frac{1}{z}$ is the lemniscate $\rho^2 = \cos 2\phi$.

[JNTU 1998S, 2007S, 2008S, Nov. 2008S, Nov. 2008, JNTU (K) Nov. 2009 (Set No. 2)]

Solution : Let $z = re^{i\theta}$, $w = \rho e^{i\phi}$. Given $w = \frac{1}{z}$

$$\Rightarrow \rho e^{i\phi} = \frac{1}{re^{i\theta}} = \frac{1}{r} e^{-i\theta} \Rightarrow \rho = \frac{1}{r}, \phi = -\theta \dots\dots (i)$$

Given hyperbola is $x^2 - y^2 = 1$

$$\Rightarrow r^2 \cos^2 \theta - r^2 \sin^2 \theta = 1 \quad (\because x = r \cos \theta, y = r \sin \theta)$$

$$\Rightarrow r^2 (\cos^2 \theta - \sin^2 \theta) = 1$$

$$\Rightarrow r^2 \cos 2\theta = 1$$

$$\Rightarrow \frac{1}{\rho^2} \cos (-2\phi) = 1 \quad [\text{Using (i)}]$$

$$\Rightarrow \rho^2 = \cos 2\phi$$

$\therefore$ The hyperbola $x^2 - y^2 = 1$ in the z -plane is mapped into lemniscate $\rho^2 = \cos 2\phi$ in the w -plane.

Example 5 : Find the effect of inversion $w = \frac{1}{z}$ on the vertical line $\operatorname{Re}(z) = a \neq 0$.

Solution : Consider the line $\operatorname{Re}(z) = a$

Let $z = a + iy$, Then

$$\begin{aligned} w = \frac{1}{z} &= \frac{1}{a + iy} = \frac{a}{a^2 + y^2} - \frac{y}{a^2 + y^2} i \\ &= u + iv \text{ (say)} \end{aligned}$$

$$\text{Then } u = \frac{a}{a^2 + y^2} \text{ and } v = \frac{-y}{a^2 + y^2}$$

$$\text{We can easily prove that } \left(u - \frac{1}{2a}\right)^2 + v^2 = \frac{1}{4a^2}$$

Thus the image of this vertical line in z -plane is a circle in $u-v$ plane with centre $\left(\frac{1}{2a}, 0\right)$

and radius = $\frac{1}{2a}$.

Example 6 : Show that the transformation $w = z + \frac{1}{z}$ converts the straight line $\arg z = a$

$\left(a < \frac{\pi}{2}\right)$ into a branch of the hyperbola of eccentricity $\sec a$. [JNTU April, Nov. 2006 (Set No. 2)]

Solution : Given transformation is $w = z + \frac{1}{z}$... (1)

Equation (1) is analytic everywhere except at a simple pole $z = 0$. Since

$$\frac{dw}{dz} = 1 - \frac{1}{z^2} = \frac{(z-1)(z+1)}{z^2}$$

w' is non-zero everywhere except at $z = \pm 1$.

Thus the mapping is conformal everywhere except at the point $z = \pm 1$

Let $z = re^{i\theta}$ and $w = u + iv$. Then equation (1) becomes

$$\begin{aligned} u + iv &= re^{i\theta} + \frac{1}{re^{i\theta}} = re^{i\theta} + \frac{1}{r} e^{-i\theta} \\ &= r(\cos \theta + i \sin \theta) + \frac{1}{r} (\cos \theta - i \sin \theta) \\ &= \left(r + \frac{1}{r}\right) \cos \theta + i \left(r - \frac{1}{r}\right) \sin \theta \end{aligned}$$

Comparing real and imaginary parts, we get

$$u = \left(r + \frac{1}{r}\right) \cos \theta \quad \dots (2)$$

$$\text{and} \quad v = \left(r - \frac{1}{r}\right) \sin \theta \quad \dots (3)$$

To find the image, we have to eliminate r . From (2) and (3), we have

$$\frac{u^2}{\cos^2 \theta} - \frac{v^2}{\sin^2 \theta} = \left(r + \frac{1}{r}\right)^2 - \left(r - \frac{1}{r}\right)^2 = 4$$

$$\text{Thus} \quad \frac{u^2}{4 \cos^2 \theta} - \frac{v^2}{4 \sin^2 \theta} = 1 \quad \dots (4)$$

Equation (4) represents a hyperbola.

$$\text{For } \theta = \arg z = a, \text{ equation (4) becomes } \frac{u^2}{4 \cos^2 a} - \frac{v^2}{4 \sin^2 a} = 1 \quad \dots (5)$$

From this we conclude that the radial lines $\theta = a \left(< \frac{\pi}{2} \right)$ (constant) in the z -plane are transformed into a family of confocal hyperbolas in the w -plane.

Here $a^2 = 4 \cos^2 a$ and $b^2 = 4 \sin^2 a$.

Eccentricity e of the hyperbola is given by

$$e^2 = 1 + \frac{b^2}{a^2}$$

$$\text{i.e. } e^2 = 1 + \frac{4 \sin^2 a}{4 \cos^2 a}$$

$$\text{i.e. } e^2 = 1 + \frac{\sin^2 a}{\cos^2 a} = \frac{\sin^2 a + \cos^2 a}{\cos^2 a} = \frac{1}{\cos^2 a} = \sec^2 a$$

$$\text{or } e = \sec a \quad \left(a < \frac{\pi}{2} \right)$$

$\therefore$ Eccentricity of the hyperbola (5) is $\sec a$.

Hence the given transformation maps the family of straight lines $\theta = a$ (constant) of the z -plane onto the family of confocal hyperbolas of the w -plane with eccentricity $\sec a$.

Example 7 : Find the image of the triangle with vertices i , $1 + i$, $1 - i$ in the z -plane under the transformation $w = 3z + 4 - 2i$ [JNTU 2002S]

$$\begin{aligned} \text{Solution : } w &= u + iv = 3z + 4 - 2i \\ &= 3(x + iy) + 4 - 2i = 3x + 4 + i(3y - 2) \end{aligned}$$

Comparing the real and imaginary parts,

$$u = 3x + 4 \text{ and } v = 3y - 2.$$

Consider the point i , ($x = 0$, $y = 1$).

Then $u = 4$, $v = 1$

$\therefore$ Image of i is $4 + i$

Similar Image of $1 + i$ is $7 + i$

and Image of $1 - i$ is $7 - i$

Thus the image of given triangle is another triangle with vertices $(4, 1)$, $(7, 1)$ and $(7, -1)$.

Example 8 : Find and plot the image of the triangular region with vertices at $(0, 0)$, $(1, 0)$, $(0, 1)$ under the transformation $w = (1 - i)z + 3$.

[JNTU 2004S, Nov. 2006, JNTU (K), (H) Nov. 2009 (Set No. 2,1)]

Solution : Given transformation is

$$w = (1 - i)z + 3$$

$$\text{i.e. } u + iv = (1 - i)(x + iy) + 3 = (x + y + 3) + i(y - x)$$

$$\Rightarrow u = x + y + 3 \text{ and } v = y - x$$

... (1)

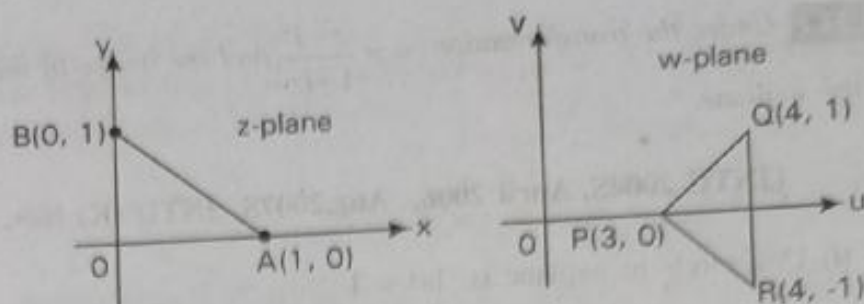


Fig. 8.16

Hence from (1) the point $(0, 0)$ is mapped to $u = 3, v = 0$ i.e. to the point $(3, 0)$ in the w -plane, the point $(1, 0)$ is mapped to $u = 4, v = -1$ i.e. to the point $(4, -1)$ in the w -plane and the point $(0, 1)$ is mapped to $u = 4, v = 1$ i.e. to the point $(4, 1)$ in the w -plane.

Example 9 : If $w = \frac{1+iz}{1-iz}$ find the image of $|z| < 1$

[JNTU 2002, 2005, Nov. 2006, JNTU (K) Nov. 2009 (Set No. 3)]

Solution : Given $w = \frac{1+iz}{1-iz}$

$$\Rightarrow w - wiz = 1 + iz \Rightarrow w - 1 = iz(1 + w) \Rightarrow z = \frac{w-1}{i(1+w)} \text{ when } |z| = \left| \frac{w-1}{i(w+1)} \right| < 1,$$

$$\Rightarrow |w-1| < |i(w+1)| = |i||w+1| = |w+1|$$

Put $w = u + iv$.

$$|u + iv - 1| < |u + iv + 1|$$

$$\Rightarrow (u-1)^2 + v^2 < (u+1)^2 + v^2$$

$$\Rightarrow u^2 + 1 - 2u + v^2 < u^2 + 1 + 2u + v^2$$

$$\Rightarrow 4u > 0 \text{ or } u > 0$$

Thus the image of the interior of the limit circle c :

$|z| = 1$ is the right-half plane in the w -plane under the transformation $w = \frac{1+iz}{1-iz}$.

Example 10 : Find the image of the unit circle $|z| = 1$ under the linear fractional transformation $w(z) = \frac{2iz - 2 - 2i}{(1-i)z - 1}$.

Solution : We know that the image of a circle is a circle or a straight line. We calculate the images of the points, $1, -1, i$ under $|z| = 1$. Writing $w(z) = \frac{2i(z-1+i)}{(1-i)z-1}$, we get

$$w(1) = -2i \text{ or } w(-1) = 2i$$

Thus, the centre of the circle must lie on the u -axis. Finally $w(i) = -2 + 4i$. We can see that the perpendicular bisector of $2i$ and $-2 + 4i$ intersect on u -axis at -4 . This is the centre of the circle.

$$\text{Thus, } |w + 4| = \sqrt{20}$$

Example 11 : Under the transformation $w = \frac{z-i}{1-iz}$, find the image of the circle (i) $|w| = 1$

(ii) $|z| = 1$ in the w -plane.

[JNTU 2004S, April 2006, Aug. 2007S, JNTU (K) Nov. 2009 (Set No. 3)]

Solution : (i) Unit circle in w -plane is $|w| = 1$

$$\text{i.e., } \left| \frac{z-i}{1-iz} \right| = 1 \quad \text{i.e., } |z-i| = |1-iz|$$

$$\text{i.e., } |(x+iy)-i| = |1-i(x+iy)| \quad \text{i.e., } |x+i(y-1)| = |(1+y)-ix|$$

$$\text{i.e., } x^2 + (y-1)^2 = (1+y)^2 + x^2 \quad \text{i.e., } x^2 + y^2 - 2y + 1 = 1 + y^2 + 2y + x^2$$

$$\text{or } 4y = 0 \quad \text{or } y = 0 \quad \text{which is real axis in the } z\text{-plane.}$$

$$(ii) \text{ Given } w = \frac{z-i}{1-iz}$$

$$\Rightarrow w - wiz = z - i \Rightarrow w + i = z(1+iw)$$

$$\Rightarrow z = \frac{w+i}{1+iw}$$

$$\text{When } |z| = 1, |w+i| = |1+iw|$$

$$\text{Putting } w = u + iv, |(u+1) + iv| = |(1-v) + iu|$$

$$\Rightarrow (u+1)^2 + v^2 = (1-v)^2 + u^2$$

$$\Rightarrow u^2 + 2u + 1 + v^2 = 1 - 2v + v^2 + u^2$$

$$\Rightarrow 2u = -2v \quad \text{or } u = -v$$

or $u + v = 0$ which is the required image of the circle.

Example 12 : Find the image of the rectangle $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}; -1 \leq y \leq 1$ under the mapping $w = \sin z$.

Solution : The rectangle has boundaries $x = \pm \frac{\pi}{2}$ and $y = \pm 1$. Hence the upper edge is mapped into the upper half of the ellipse $\frac{u^2}{\cosh^2 y} + \frac{v^2}{\sinh^2 y} = 1$ and the lower edge is mapped on to the lower half of the above ellipse. The left edge is mapped onto the segment $-\cosh 1 \leq u \leq -1$ and the right-edge is mapped onto the segment $|1 \leq u \leq \cosh u|$ of the u -axis.

Also any two points $\pm y_0$ of these edges have the same image, $\cosh y_0$ (left edge) or $\cosh y_0$ (right edge) on u -axis.

(Refer the mapping $w = \sin z$)

Example 13 : Find the image of the rectangle $R: -\pi < x < \pi, \frac{1}{2} < y < 1$ under the transformation $w = \sin z$.
[JNTU 2003 (Set No. 1)]

Solution : We have $w = u + iv = \sin z = \sin(x + iy)$

$$\begin{aligned} \text{i.e. } u + iv &= \sin x \cdot \cos iy + \cos x \sin iy \\ &= \sin x \cdot \cosh y + i \cos x \cdot \sinh y \\ \therefore u &= \sin x \cosh y \text{ and } v = \cos x \sinh y \end{aligned}$$

$$(\because \cos ix = \cosh x \text{ and } \sin ix = i \sinh x)$$

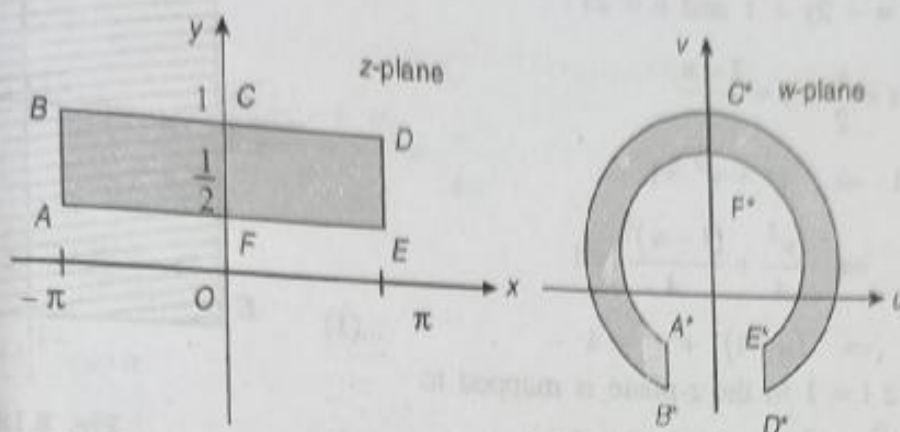


Fig. 8.17

The rectangle has boundaries

$$x = \pm \pi \text{ and } y = \frac{1}{2}, y = 1.$$

Hence the upper edge BCD i.e. $y = 1$ is mapped into the outer ellipse ($B^*C^*D^*$)

$$\frac{u^2}{\cosh^2 1} + \frac{v^2}{\sinh^2 1} = 1$$

with semi-axes $\cosh 1$ and $\sinh 1$.

The lower edge AFE i.e. $y = \frac{1}{2}$ is mapped onto the inner ellipse ($A^*F^*E^*$)

$$\frac{u^2}{\cosh^2 \frac{1}{2}} + \frac{v^2}{\sinh^2 \frac{1}{2}} = 1$$

with semi-axes $\cosh \frac{1}{2} = 1.13$ and $\sinh \frac{1}{2} = 0.52$

The left edge $AB: x = -\pi, -\frac{1}{2} < y < 1$ is mapped onto $u = 0$ and $-\sinh \frac{1}{2} < v < -\sinh 1$ given by A^*B^* and the right edge $DE: x = \pi, \frac{1}{2} < y < 1$ is mapped onto $u = 0, v = -\sinh y$ given by D^*E^* .

Example 14 : Plot the image $1 < |z| < 2$ under the transformation $w = 2iz + 1$. [JNTU 1995S]

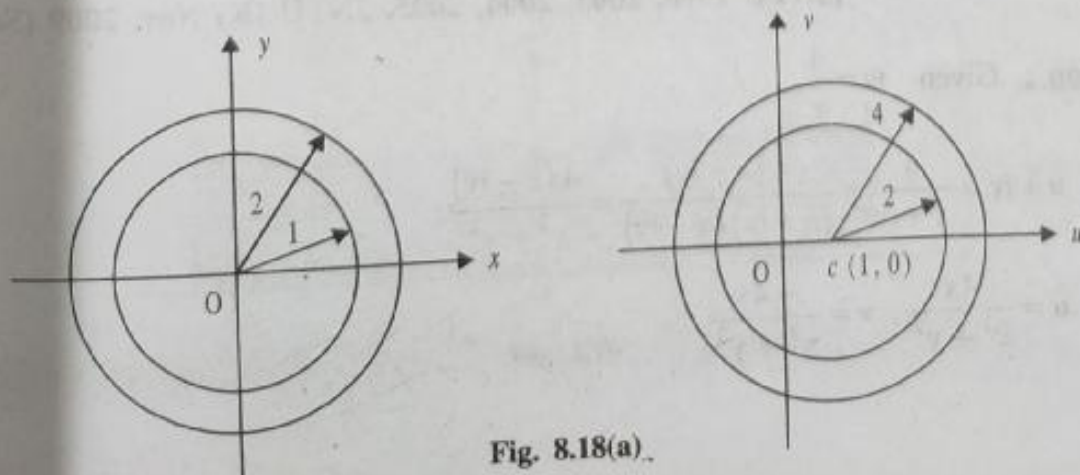


Fig. 8.18(a).

Solution : Given $w = 2iz + 1$.

$$\Rightarrow u + iv = 2i(x + iy) + 1$$

$$\therefore u = -2y + 1 \text{ and } v = 2x$$

$$\Rightarrow x = \frac{v}{2}, y = \frac{1-u}{2}$$

$$|z|=1 \Rightarrow x^2 + y^2 = 1$$

$$\Rightarrow \frac{v^2}{4} + \frac{(1-u)^2}{4} = 1$$

$$\Rightarrow (u-1)^2 + v^2 = 4 \quad \dots(1)$$

i.e., the circle $|z|=1$ in the z -plane is mapped to the circle $(u-1)^2 + v^2 = 4$ in the w -plane.

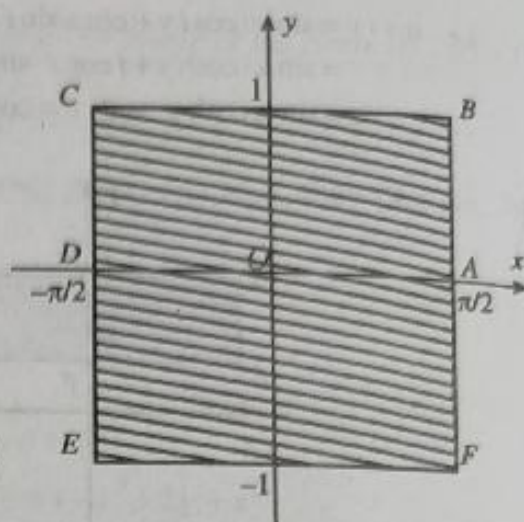


Fig. 8.18(b)

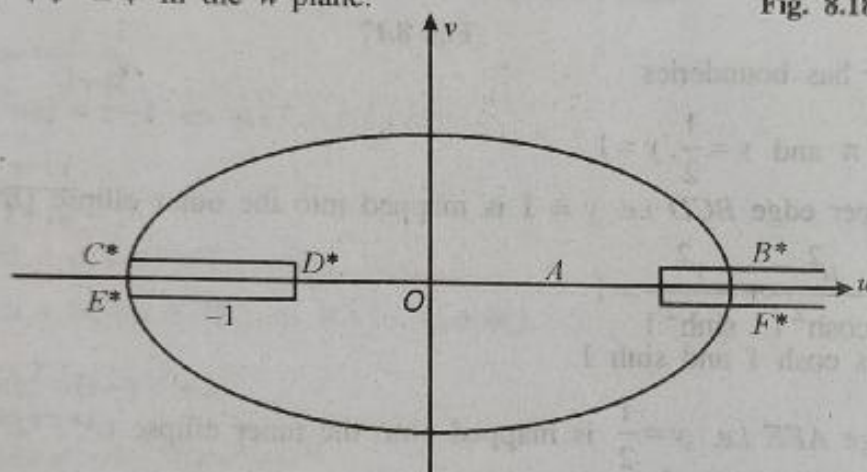


Fig. 8.18(c)

Similarly $|z|=2 \Rightarrow x^2 + y^2 = 4$

$$\Rightarrow \frac{v^2}{4} + \frac{(1-u)^2}{4} = 4 \Rightarrow (u-1)^2 + v^2 = 16 \quad \dots(2)$$

which is again a circle in the w -plane with centre $(1, 0)$ and radius $= 4$.

Hence, the region between the two concentric circles $|z|=1$ and $|z|=2$ is mapped onto the annular region between the two concentric circles given by (1) and (2) in the w -plane.

Example 15 : Show that the function $w = \frac{4}{z}$ transforms the straight line $x = c$ in the z -plane into a circle in the w -plane.

[JNTU 1998, 2003, 2004, 2005, JNTU (K) Nov. 2009 (Set No. 4)]

Solution : Given $w = \frac{4}{z}$

$$\Rightarrow u + iv = \frac{4}{x + iy} = \frac{4(x - iy)}{(x + iy)(x - iy)} = \frac{4(x - iy)}{x^2 + y^2}$$

$$\therefore u = \frac{4x}{x^2 + y^2}, \quad v = \frac{-4y}{x^2 + y^2}$$

$$\text{When } x = c, \quad u = \frac{4c}{c^2 + y^2}, \quad \dots(1), \quad v = \frac{-4y}{c^2 + y^2} \quad \dots(2)$$

$$\Rightarrow y^2 = \frac{4c - c^2 u}{u}$$

$$\text{Now } v^2 = \frac{16y^2}{(c^2 + y^2)^2} = \frac{16(4c - c^2 u)}{u} \times \frac{u^2}{16c^2}$$

$$\therefore v^2 = \frac{(4 - cu)u}{c}$$

$$\Rightarrow cv^2 = (4 - cu)u$$

$$\Rightarrow u^2 + v^2 - \frac{4}{c}u = 0 \text{ which is a circle in the } w\text{-plane.}$$

Example 16 : Find the image of the semi-infinite strip $x \geq 0, 0 \leq y \leq \pi$ under the mapping $w = \cosh z$.

Solution : We set $w = u + iv$. Since $\cosh 0 = 1$, the point $z = 0$ is mapped onto $w = 1$.

For real $z = x \geq 0$, $\cosh z$ is real and increases with increasing x in a monotone fashion, starting from 1. Hence the positive x -axis is mapped onto the portion $u \geq 1$ of the u -axis.

For pure imaginary $z = iy$ we have $\cosh iy = \cos y$. Hence the left boundary of the strip is mapped onto the segment $1 \geq u \geq -1$ of the u -axis, the point $z = \pi i$ corresponding to mapped onto the segment $1 \geq u \geq -1$ of the u -axis, the point $z = \pi i$ corresponding to $w = \cosh i\pi = \cos \pi = -1$. On the upper boundary of the strip $y = \pi$, and since $\sin \pi = 0$, $\cos \pi = -1$, it follows that this part of the boundary is onto the portion $u \leq -1$ of the u -axis. Hence the boundary of the strip is mapped onto the u -axis. It is not difficult to see that the interior of the strip is mapped onto the upper half of the w -plane, and the mapping is one-to-one.

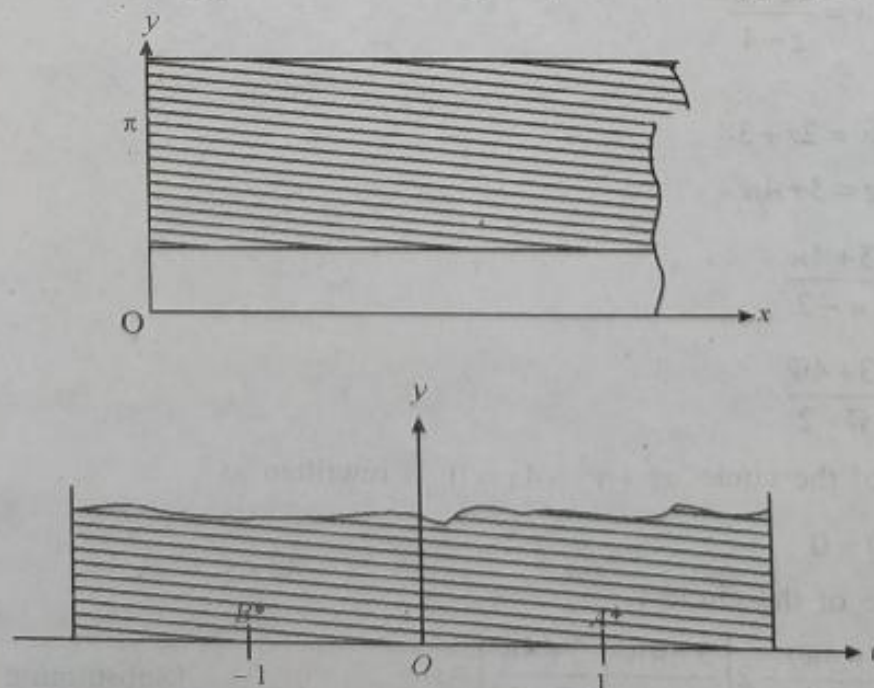


Fig. 8.19

Example 17 : Find the region in the w -plane in which the rectangle bounded by the lines $x = 0$, $y = 0$, $x = 2$ and $y = 1$ is mapped under the transformation $w = z + (2 + 3i)$

[JNTU (K) Nov. 2009 (Set No. 4)]

Solution : The given transformation is $w = z + (2 + 3i)$... (1)

Let $w = u + iv$ and $z = x + iy$. Then (1) becomes $u + iv = (x + 2) + i(y + 3)$

$\Rightarrow u = x + 2$ and $v = y + 3$... (2)

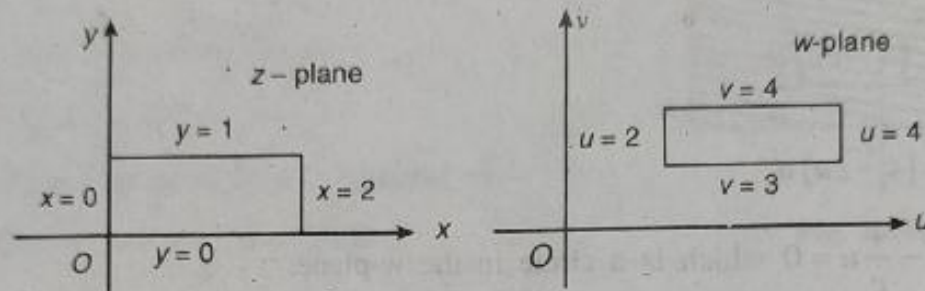


Fig. 8.20

Now, from (2),

when $x = 0$, $u = 2$

$y = 0$, $v = 3$

and when $x = 2$, $u = 4$

$y = 1$, $v = 4$

Hence the given rectangle in the z -plane is mapped into the rectangle bounded by the lines $u = 2$, $v = 3$, $u = 4$ and $v = 4$, in the w -plane as shown in the figure.

Example 18 : Show that the transformation $w = \frac{2z+3}{z-4}$ changes the circle $x^2 + y^2 - 4x = 0$ into the straight line $4u + 3 = 0$

[JNTU 2003, 2007S, 2008S, Nov. 2008S (Set No. 2)]

Solution : Given transformation is

$$w = \frac{2z+3}{z-4}$$

Solving for z ,

$$wz - 4w = 2z + 3$$

$$\Rightarrow (w-2)z = 3+4w$$

$$\therefore z = \frac{3+4w}{w-2} \quad \dots (1)$$

$$\text{Hence } \bar{z} = \frac{3+4\bar{w}}{\bar{w}-2} \quad \dots (2)$$

The equation of the circle $x^2 + y^2 - 4x = 0$ is rewritten as

$$z\bar{z} - 2(z + \bar{z}) = 0 \quad \dots (3)$$

Thus the image of the circle is

$$\frac{(3+4w)(3+4\bar{w})}{(w-2)(\bar{w}-2)} - 2\left[\frac{3+4w}{w-2} + \frac{3+4\bar{w}}{\bar{w}-2}\right] = 0$$

[Substituting (1) & (2) in (3)]

$$\text{i.e. } (3+4w)(3+4\bar{w}) - 2[(3+4w)(\bar{w}-2) + (w-2)(3+4\bar{w})] = 0$$

$$\text{i.e. } 9+16w\bar{w}+12(w+\bar{w}) - 2[(-6+4w\bar{w}+3\bar{w}-8w) + (4w\bar{w}+3w-6-8\bar{w})] = 0$$

$$\text{i.e. } 22(w+\bar{w})+33=0$$

$$\text{or } 2(w+\bar{w})+3=0$$

$$\text{or } 4u+3=0$$

which is a straight line in the w -plane.

Example 19 : Show that the relation $w = \frac{5-4z}{4z-2}$ transform the circle $|z| = 1$ into a circle of radius unity in the w -plane. [JNTU 2003, 2005 (Set No. 3)]

Solution : Given transformation is $w = \frac{5-4z}{4z-2}$

$$\text{Solving for } z, \quad 4wz - 2w = 5 - 4z$$

$$\Rightarrow 4z(w+1) = 5+2w$$

$$\Rightarrow z = \frac{5+2w}{4(w+1)} \quad \therefore \bar{z} = \frac{5+2\bar{w}}{4(\bar{w}+1)}$$

$$\text{Now } |z| = 1 \text{ or } z\bar{z} = 1. \text{ This corresponds to } \left[\frac{5+2w}{4(w+1)} \right] \left[\frac{5+2\bar{w}}{4(\bar{w}+1)} \right] = 1$$

$$\text{i.e. } (5+2w)(5+2\bar{w}) = 16(w+1)(\bar{w}+1)$$

$$\text{i.e. } 25+10w\bar{w}+10(w+\bar{w})+4w\bar{w} = 16(w\bar{w}+w+\bar{w}+1)$$

$$\text{i.e. } 12w\bar{w}+6w+6\bar{w}-9=0$$

$$\text{i.e. } w\bar{w} + \frac{w}{2} + \frac{\bar{w}}{2} - \frac{3}{4} = 0$$

$$\text{or } w\bar{w} + \frac{1}{2}(w+\bar{w}) - \frac{3}{4} = 0 \quad \dots (1)$$

$$\text{But } w = u+iv, \bar{w} = u-iv, w\bar{w} = u^2+v^2, w+\bar{w} = 2u$$

$$\text{Hence (1)} \Rightarrow u^2+v^2+u-\frac{3}{4}=0$$

$$\text{i.e. } \left(u + \frac{1}{2}\right)^2 + v^2 = \frac{3}{4} + \frac{1}{4} = 1$$

This is a circle with centre at $\left(-\frac{1}{2}, 0\right)$ and of radius one in w -plane.

Example 20 : Show that the transformation $w = \frac{z-i}{z+i}$ maps the real axis in the z -plane into the unit circle $|w|=1$ in the w -plane [JNTU 2003, Nov. 2008S (Set No. 4)]

Solution : Unit circle in w -plane is

$$|w|=1$$

$$\Rightarrow \left| \frac{z-i}{z+i} \right| = 1 \Rightarrow |z-i| = |z+i|$$

$$\text{i.e. } |x+i(y-1)| = |x+i(y+1)|$$

$$\text{i.e. } x^2 + (y-1)^2 = x^2 + (y+1)^2 \quad \text{i.e. } x^2 + y^2 - 2y + 1 = x^2 + y^2 + 2y + 1$$

$$4y = 0 \quad \text{or} \quad y = 0$$

which is real axis in the z -plane.

Example 21 : Find the image of the triangular region in the z -plane bounded by the lines $x = 0$, $y = 0$ and $x + y = 1$ under the transformation $w = 2z$. [JNTU 2003S (Set No. 2)]

Solution : The transformation is $w = 2z$ so that $u + iv = 2(x + iy)$

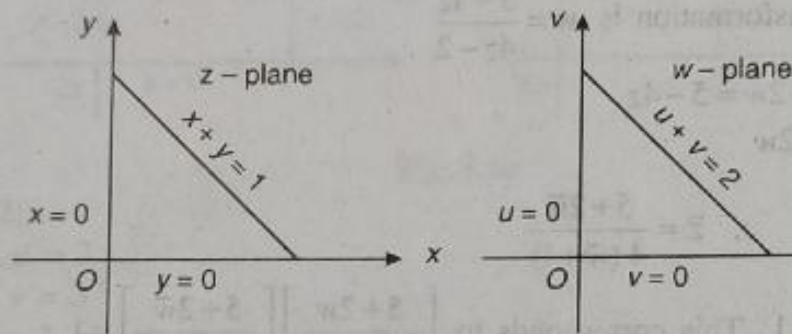


Fig. 8.21

$$\text{Hence } u = 2x \text{ and } v = 2y \quad \dots (1)$$

$$\text{when } x = 0, u = 0 \quad \dots (2)$$

$$\text{and when } y = 0, v = 0$$

$$(1) + (2) \text{ gives } u + v = 2(x + y) = 2(1) \quad (\because x + y = 1)$$

$$\text{or} \quad u + v = 2$$

Thus we have shown that the given map $w = 2z$ transforms the triangle in z -plane onto the triangle in the w -plane bounded by $u = 0$, $v = 0$ and $u + v = 2$.

Example 22 : Find the image of the triangle with vertices at i , $1 + i$, $1 - i$ in the z -plane under the transformation $w = 3z + 4 - 2i$. [JNTU 2003S (Set No. 3)]

Solution : The given transformation is $w = 3z + 4 - 2i$

$$\text{or} \quad u + iv = 3(x + iy) + 4 - 2i = (3x + 4) + i(3y - 2)$$

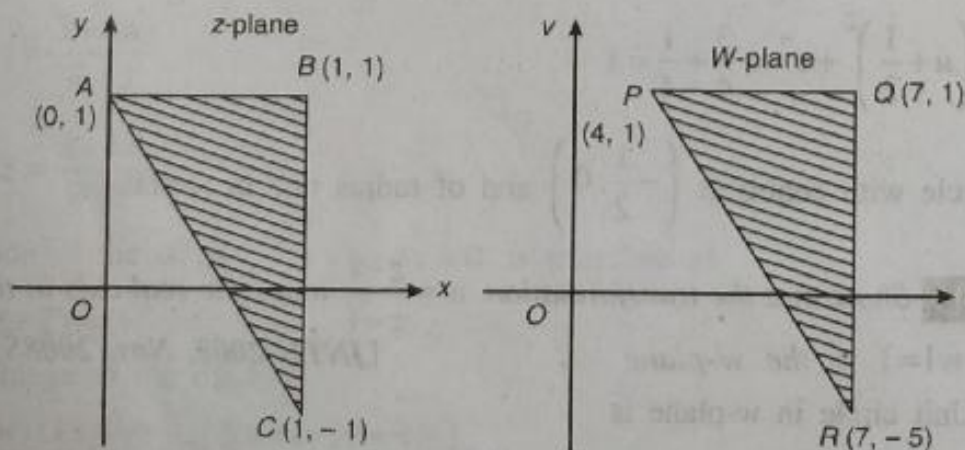


Fig. 8.22

Equating real and imaginary parts,

$$u = 3x + 4 \quad \dots (1)$$

$$v = 3y - 2 \quad \dots (2)$$

Thus from (1) and (2), the point (0, 1) in the z -plane is mapped into the point (4, 1) in the w -plane, the point (1, 1) in the z -plane is mapped into the point (7, 1) in the w -plane and the point (1, -1) in z -plane is mapped into the point (7, -5) in w -plane.

Hence the triangle ABC in z -plane is transformed to the triangle PQR in w -plane under the given transformation.

Example 23 : By the transformation $w = z^2$, show that the circles $|z - a| = c$ (a, c being positive and unequal) in the z -plane correspond to the limacons in the w -plane

[JNTU 2001S, 2008S (Set No. 2)]

Solution : The transformation is $w = z^2$... (1)

Given circle is $|z - a| = c$ so that $z - a = c e^{i\theta}$

$$\text{or } z = a + c e^{i\theta}, \quad \dots (2)$$

for any point on the circle.

$$\therefore w = z^2 = (a + c e^{i\theta})^2 \quad [\text{From (2)}]$$

$$= a^2 + 2ac e^{i\theta} + c^2 e^{2i\theta}$$

$$\text{or } w - a^2 + c^2 = c^2 + 2ac e^{i\theta} + c^2 e^{2i\theta}$$

$$= c e^{i\theta} [c e^{i\theta} + c e^{-i\theta} + 2a]$$

$$= c e^{i\theta} (2c \cos \theta + 2a)$$

$$\text{or } w - (a^2 - c^2) = 2c e^{i\theta} (a + c \cos \theta) \quad \dots (3)$$

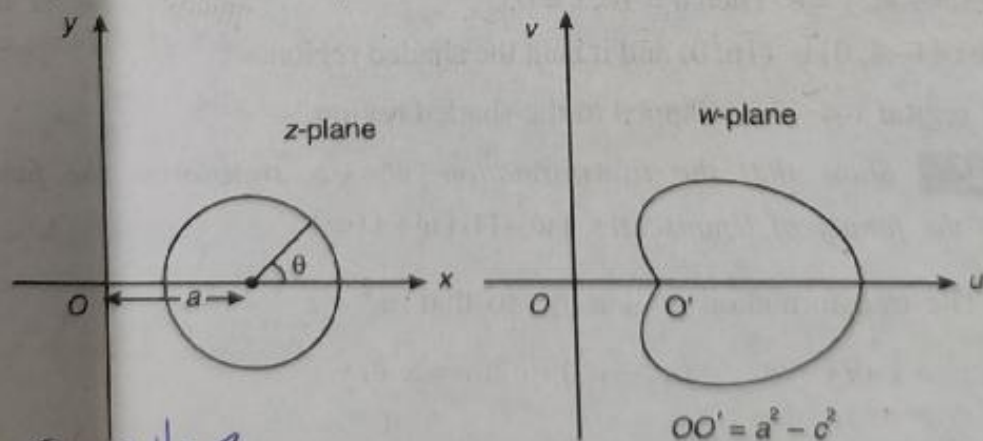


Fig. 8.23

$$\text{Let } |w - (a^2 - c^2)| = R$$

Putting $w - (a^2 - c^2) = R e^{i\phi}$ i.e. by taking the pole at $w = a^2 - c^2$ in the w -plane, (3) takes the form $R e^{i\phi} = 2c e^{i\theta} (a + c \cos \theta)$

Equating modulus and amplitude on both sides, we have

$$R = 2c (a + c \cos \theta), \phi = \theta$$

or $R = 2c(a + c \cos \phi)$ which is a limaçon in w -plane.

Thus the circle $|z - a| = c$ corresponds to a limaçon in w -plane.

Deduction. If $a = c$, then the circle $|z - a| = c$ becomes $|z - a| = a$ which is a circle in z -plane of radius a and centre at $(a, 0)$. Also $R = 2c(a + c \cos \phi)$ becomes $R = 2a^2(1 + \cos \phi)$ which is a cardioid.

Example 24 : Find the image of the domain in the z -plane to the left of the line $x = -3$ under the transformation $w = z^2$. [JNTU 2006, 2008S (Set No. 4)]

Solution : We have

$$w = u + iv = z^2 = (x + iy)^2 = x^2 - y^2 + i2xy$$

Equating real and imaginary parts, we get

$$u = x^2 - y^2 \quad \dots (1) \text{ and}$$

$$v = 2xy \quad \dots (2)$$

If $x = -3$, then (1) and (2) becomes $u = 9 - y^2 \quad \dots (3)$

$$\text{and } v = -6y \Rightarrow y = -\frac{v}{6} \quad \dots (4)$$

$$\text{Eliminating } y, \text{ we get } u = 9 - \left(-\frac{v}{6}\right)^2 = 9 - \frac{v^2}{36}$$

$$\text{or } \frac{v^2}{36} = 9 - u \quad \dots (5)$$

If $u = 0$, then (5) becomes $\frac{v^2}{36} = 9$ or $v = \pm 18$

$$v = 0, \text{ then } u = 9$$

$\therefore$ The line $x = -3$ is mapped on to the parabola $v^2 = 36(9 - u)$.

Now take $x = -4, y = 0$. Then $u = 16, v = 0$.

The image of $(-4, 0)$ is $(16, 0)$ and it is in the shaded region.

Hence the region $x < -3$ is mapped to the shaded region.

Example 25 : Show that the transformation $w = \sqrt{z}$ transforms the family of circles $|z - 1| = \lambda$ into the family of lemniscates $|w - 1| \cdot |w + 1| = \lambda$.

Solution : The transformation is $w = \sqrt{z}$ so that $w^2 = z$

$$\text{or } (u + iv)^2 = x + iy \quad \text{or } (u^2 - v^2) + i2uv = x + iy$$

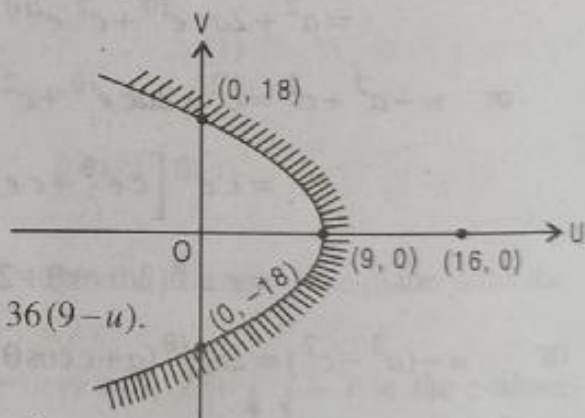
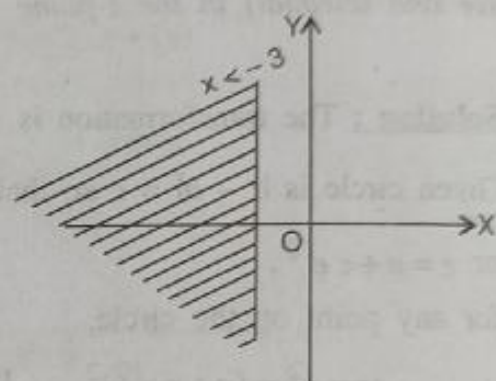
$$\Rightarrow u^2 - v^2 = x$$

$$\text{and } 2uv = y$$

The circle is $|z - 1| = \lambda$

$$\text{i.e. } |(x - 1) + iy| = \lambda$$

$$\text{i.e. } (x - 1)^2 + y^2 = \lambda^2$$



$$\text{or } x^2 + y^2 + 1 - 2x = \lambda^2 \quad \dots (3)$$

Substituting the values of x and y from (1) and (2) in (3), we get

$$(u^2 - v^2)^2 + 4u^2v^2 + 1 - 2(u^2 + v^2) = \lambda^2 \quad \dots (4)$$

$$\text{Now } |\omega - 1|^2 \cdot |\omega + 1|^2 = |\omega^2 - 1|^2$$

$$= |(u + iv)^2 - 1|^2 = |(u^2 - v^2 - 1) + i2uv|^2 \\ = (u^2 - v^2 - 1)^2 + 4u^2v^2$$

$$\text{or } |\omega - 1|^2 \cdot |\omega + 1|^2 = (u^2 - v^2)^2 + 1 - 2(u^2 - v^2) + 4u^2v^2 = \lambda^2 \quad [\text{From (4)}]$$

$$\text{or } |\omega - 1| \cdot |\omega + 1| = \lambda \quad \dots (5)$$

Thus we have shown that the given map $\omega = \sqrt{z}$ transforms the circles $|z - 1| = \lambda$ into family of lemniscates $|\omega - 1| |\omega + 1| = \lambda$ i.e. equation (3) onto the curve (5).

Note. When $\lambda = 1$, (5) becomes $|\omega - 1| \cdot |\omega + 1| = 1$

$$\text{or } (u^2 - v^2)^2 + 1 - 2(u^2 - v^2) + 4u^2v^2 = 1 \quad [\text{using (4)}]$$

$$\text{or } (u^2 + v^2) - 2(u^2 - v^2) = 0.$$

$$\text{Its polar equivalent is } R^4 - 2R^2 \cos 2\phi = 0$$

$$\text{or } R^2 = 2 \cos 2\phi$$

which is a lemniscate.

Hence the result follows.

Example 26 : Show that the transformation $w = z^2$ maps the circle $|z - 1| = 1$ into the cardioid $r = 2(1 + \cos \theta)$ where $w = r e^{i\theta}$ in the w -plane. [JNTU 2005S(Set No. 2)]

Solution : The transformation is $w = z^2$

$$\text{Given circle is } |z - 1| = 1 \text{ so that } z - 1 = e^{i\phi} \text{ or } z = 1 + e^{i\phi} \quad \dots (1)$$

for any point on the circle.

$$\therefore w = z^2 = (1 + e^{i\phi})^2 \quad [\text{From (1)}] = 1 + e^{i2\phi} + 2e^{i\phi} = e^{i\phi} [e^{-i\phi} + e^{i\phi} + 2] \\ = e^{i\phi} \left[e^{\frac{i\phi}{2}} + e^{\frac{-i\phi}{2}} \right]^2 = e^{i\phi} \left(2 \cos \frac{\theta}{2} \right)^2 = 4 e^{i\phi} \cos^2(\theta/2) = e^{i\phi} [e^{-i\phi/2} + e^{i\phi/2}]^2$$

$$\text{or } r e^{i\theta} = 4 e^{i\phi} \cos^2 \frac{\theta}{2}$$

$$\Rightarrow r = 4 \cos^2 \frac{\theta}{2} \text{ and } \theta = \phi$$

$$\therefore r = 2 \left(2 \cos^2 \frac{\theta}{2} \right) = 2(1 + \cos \theta)$$

which is a cardioid in w -plane.

Aliter.

$$\text{Let } z = r_1 e^{i\phi} \text{ and } w = r e^{i\theta}$$

$$\text{Given transformation is } w = z^2$$

$$\text{i.e. } r e^{i\theta} = r_1^2 e^{i2\phi} \Rightarrow r = r_1^2 \text{ and } \theta = 2\phi \quad \dots (1)$$

Given circle is $|z - 1| = 1$

$$\text{i.e. } |x + iy - 1| = \text{i.e. } |(x-1) + iy| = 1$$

$$\text{or } (x-1)^2 + y^2 = 1 \text{ or } x^2 + y^2 - 2x = 0$$

$$\text{or } r_1^2 - 2r_1 \cos \phi = 0 \quad \because (x = r_1 \cos \phi, y = r_1 \sin \phi)$$

$$\text{or } r_1 = 2 \cos \phi$$

$$\therefore r_1^2 = 4 \cos^2 \phi = 2(1 + \cos 2\phi)$$

$$\text{or } r = 2(1 + \cos \theta) \text{ [by (1)]}$$

Hence the circle $|z - 1| = 1$ in the z -plane transforms into the cardioid $r = 2(1 + \cos \theta)$ in the w -plane.

Example 27 : Find and plot the rectangular region $0 \leq x \leq 1, 0 \leq y \leq 2$ under the transformation $w = \sqrt{2} e^{i\pi/4} z + (1 - 2i)$. [JNTU April 2006, Nov. 2008S, June 2009S (Set No. 2)]

Solution : Given transformation is

$$w = \sqrt{2} e^{i\pi/4} z + (1 - 2i)$$

$$\text{i.e. } u + iv = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) (x + iy) + (1 - 2i)$$

$$= \sqrt{2} \left(\frac{1}{\sqrt{2}} + i \cdot \frac{1}{\sqrt{2}} \right) (x + iy) + (1 - 2i)$$

$$= (1 + i)(x + iy) + (1 - 2i)$$

$$= (x - y) + i(x + y) + (1 - 2i)$$

$$= (x - y + 1) + i(x + y - 2)$$

$$\Rightarrow u = x - y + 1 \text{ and } v = x + y - 2 \quad \dots (1)$$

Putting $x = 0$ in (1), $u = -y + 1, v = y - 2$ so that $v = -u - 1$

Putting $x = 1$ in (1), $u = 2 - y, v = y - 1$ so that $v = 1 - u$

Putting $y = 0$ in (1), $u = x + 1, v = x - 2$ so that $v = u - 3$

Putting $y = 2$ in (1), $u = x - 1, v = x$ so that $v = u + 1$

Thus the region is a rectangle bounded by the lines $v = -u - 1, v = 1 - u, v = u - 3$ and $v = u + 1$.

The two regions are shown below.

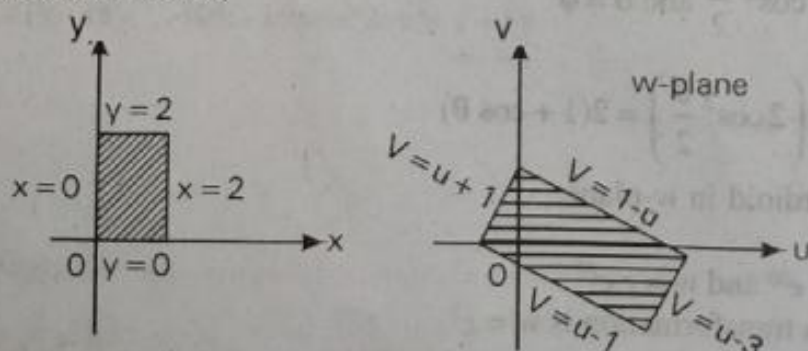


Fig. 8.24

✓ **Example 28 :** Find the image of the line $x = 4$ in z -plane under the transformation $w = z^2$.

[JNTU April 2006 (Set No. 2)]

Solution : Given transformation is

$$w = z^2$$

i.e. $u + iv = (x + iy)^2 = x^2 - y^2 + i2xy$

Comparing real and imaginary parts, we get

$$u = x^2 - y^2 \quad \dots (1)$$

and $v = 2xy \quad \dots (2)$

To get the image of the line $x = 4$, put $x = 4$ in the equation (1) and (2)

$$\therefore u = 16 - y^2 \quad \dots (3) \text{ and } v = 8y \quad \dots (4)$$

To get the relation in u and v , we have to eliminate y .

From (4), $y = \frac{v}{8}$

Putting the value of y in (3), we get

$$u = 16 - \frac{v^2}{64} \quad \text{or} \quad 64u = 1024 - v^2$$

$$\text{or } v^2 = -64(u - 16) = -4(4^2)(u - 4^2)$$

This represents a parabola in w -plane with vertex at $(16, 0)$ and focus at the origin and symmetrical about the real line

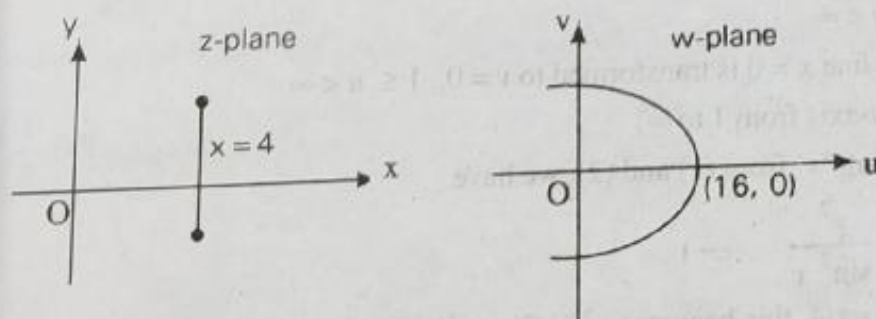


Fig. 8.25

4 **Example 29 :** Find the image of the region in the z -plane between the lines $y = 0$ and $y = \frac{\pi}{2}$ under the transformation $w = e^z$.

[JNTU April 2006, JNTU (H) Nov. 2009 (Set No. 4)]

Solution : Let $z = x + iy$ and $w = R e^{i\phi}$

Given transformation is $w = e^z$

$$\therefore R e^{i\phi} = e^{x+iy} = e^x \cdot e^{iy} \quad \dots (2)$$

Equating, $R = e^x \quad \dots (1) \text{ and } \phi = y$

If $y = 0$ then $\phi = 0$ represents radial line making an angle of zero radians with the x -axis.

If $y = \frac{\pi}{2}$ then $\phi = \frac{\pi}{2}$ represents radial line making an angle of $\frac{\pi}{2}$ radians with the x -axis.

Thus when $R = e^x$ increases from 0 to ∞ and monotonically as x takes values from $-\infty$ to ∞ , the line $y = \pi/2$ in z -plane is mapped onto the ray $\phi = \frac{\pi}{2}$ excluding the origin in w -plane.

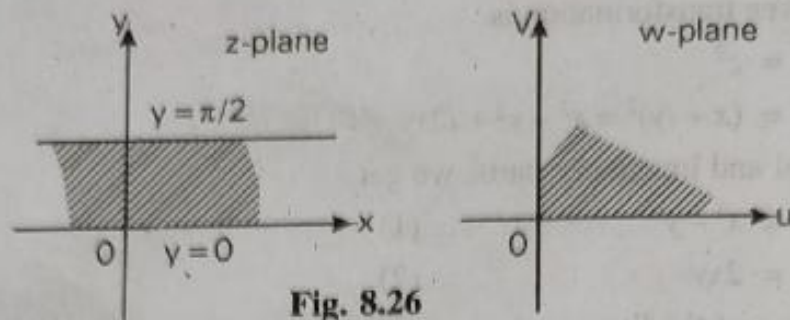


Fig. 8.26

Hence the infinite strip bounded by the lines $y = 0$ and $y = \frac{\pi}{2}$ is mapped onto the upper quadrant of w -plane.

Example 30 : The image of the infinite strip bounded by $x = 0$ and $x = \pi/4$ under the transformation $w = \cos z$. [JNTU A 2005]

Solution : Consider $w = \cos z$

$$\Rightarrow u + iv = \cos(x + iy) = \cos x \cos iy - \sin x \sin iy \\ = \cos x \cdot \cosh y = i \sin x \cdot \sinh y$$

Comparing real and imaginary parts

$$\therefore u = \cos x \cdot \cosh y \quad \dots (1), \quad v = -\sin x \cdot \sinh y \quad \dots (2)$$

When $x = 0$, $u = \cosh y$, and $v = 0$

$$\text{When } -\infty < y < \infty \Rightarrow 1 \leq \cosh y < \infty$$

$$\text{i.e., } 1 \leq u < \infty$$

Thus the line $x = 0$ is transformed to $v = 0$, $1 \leq u < \infty$

(i.e. the u -axis from 1 to ∞)

Eliminating 'y' from (1) and (2), we have

$$\frac{u^2}{\cos^2 x} - \frac{v^2}{\sin^2 x} = 1$$

$$\text{when } x = \pi/4, \text{ this becomes } u^2/(1/2) - v^2/(1/2) = 1$$

$$\text{or } u^2 - v^2 = 1/2$$

Thus the line $x = \pi/4$ is transformed to the rectangular hyperbola $u^2 - v^2 = \frac{1}{2}$ in the w -plane under the transformation $w = \cos z$.

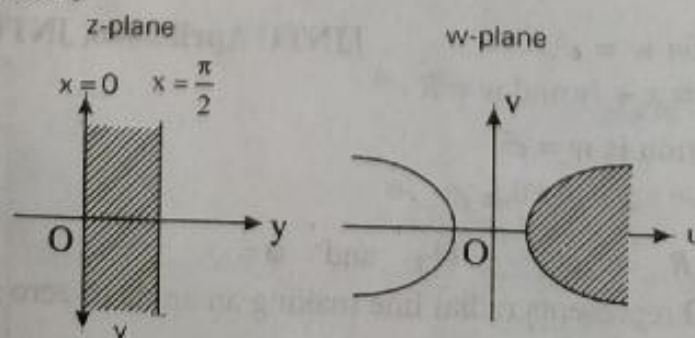


Fig. 8.27

The infinite strip bounded by $x = 0$ and $x = \pi/4$ in the z -plane is transformed to the region between the u -axis from 1 to ∞ and the rectangular hyperbola $u^2 - v^2 = \frac{1}{2}$.

Example 31 : Find and plot the image of the regions

- (i) $x > 1$ (ii) $y > 0$ (iii) $0 < y < (1/2)$

[JNTU Nov. 2008 (Set No. 2)]

under the transformation $w = 1/z$.

Solution : (i) The given transformation is $w = 1/z$

Let $w = u + iv$ and $z = x + iy$

$$w = \frac{1}{z} \Rightarrow z = \frac{1}{w} = \frac{1}{u + iv}$$

$$\therefore x + iy = \frac{u - iv}{u^2 + v^2} \Rightarrow x = \frac{u}{u^2 + v^2}; y = \frac{-v}{u^2 + v^2}$$

Consider the region $x > 1$

$$\therefore \frac{u}{u^2 + v^2} > 1 \Rightarrow u^2 + v^2 - u < 1$$

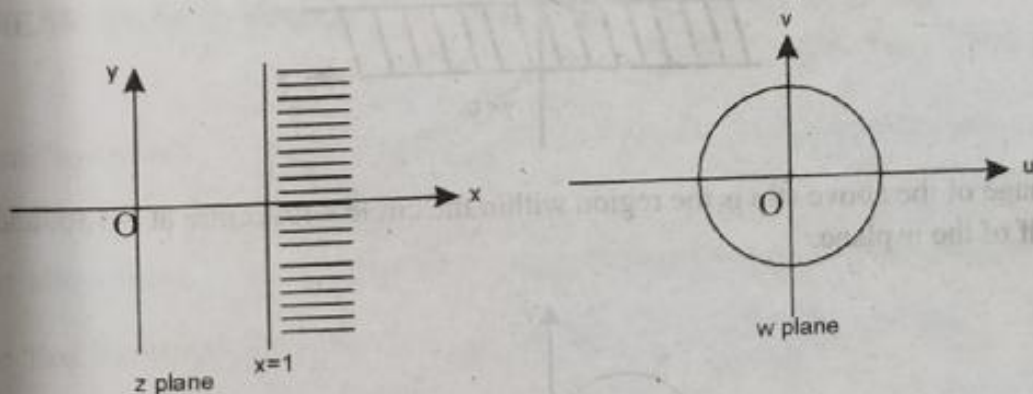
$$\Rightarrow u^2 + v^2 - u - 1 < 0$$

But $u^2 + v^2 - u - 1 = 0$ is a circle in the (u, v) plane with centre at $\left(\frac{1}{2}, 0\right)$ and

$$\text{radius} = \sqrt{\frac{1}{4} + 1} = \frac{\sqrt{5}}{2}$$

$\therefore u^2 + v^2 - u - 1 < 0$ in the region in the (u, v) plane which is within the circle with

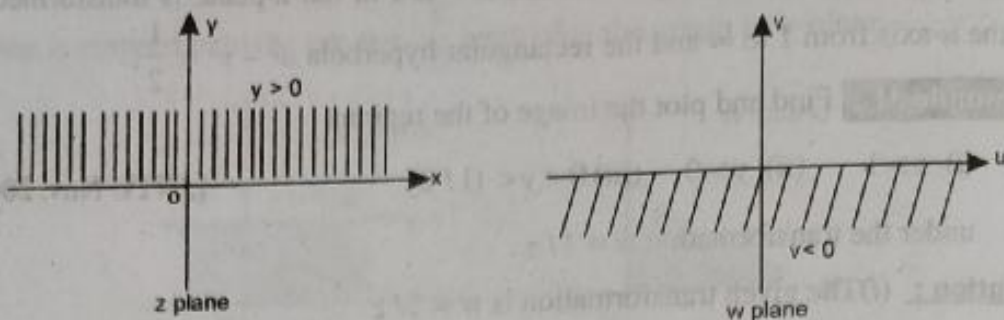
centre $\left(\frac{1}{2}, 0\right)$ and radius $\frac{\sqrt{5}}{2}$.



(ii) Consider the region $y > 0$

$$y > 0 \Rightarrow \frac{-v}{u^2 + v^2} > 0$$

$$\therefore -v > 0 \therefore v < 0.$$



$\therefore$ The upper half of the z plane is mapped onto the lower half of the w plane.

(iii) Consider the region $0 < y < \frac{1}{2}$

$$y > 0 \Rightarrow v < 0$$

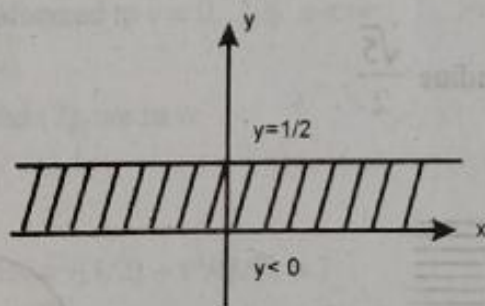
$$y < \frac{1}{2} \Rightarrow \frac{-v}{u^2 + v^2} < \frac{1}{2}$$

$$\Rightarrow u^2 + v^2 > -2v$$

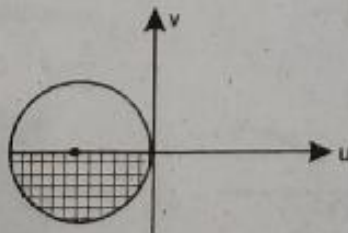
$$\Rightarrow u^2 + v^2 > +2v < 0$$

$u^2 + v^2 + 2v = 0$ is a circle in (u, v) plane with centre at $(-1, 0)$ and radius 1 unit.

The region $0 < y < \frac{1}{2}$ in the z -plane is the strip between the lines $y = 0$ and $y = \frac{1}{2}$ (lines not included).



The image of the above strip is the region within the circle with centre at $(-1, 0)$, and radius 1 in the lower half of the w plane.



Example 32 : Prove that the transformation $w = \sin z$ maps the families of lines $x = a$ and $x = b$ into two families of confocal central conics.

Solution : We have $u + iv = \sin(x + iy) = \sin x \cosh y + i \cos x \sinh y$
 $= \sin x \cosh y + i \cos x \sinh y$

$$\Rightarrow u = \sin x \cosh y \quad \dots (1)$$

$$v = \cos x \sinh y \quad \dots (2)$$

Since $\cosh^2 y - \sinh^2 y = 1$, eliminating y from (1) and (2), we get

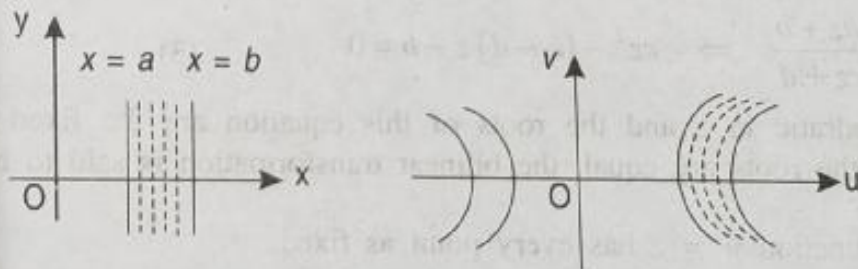
$$\frac{u^2}{\sin^2 x} - \frac{v^2}{\cos^2 x} = 1 \quad \dots (3)$$

When $x = a$, the equation (3) gives

$$\frac{u^2}{\sin^2 a} - \frac{v^2}{\cos^2 a} = 1 \quad \dots (4)$$

(4) represents a system of hyperbolas having foci at $(1, 0)$ and $(-1, 0)$.

Similarly, when $x = b$, the equation (3) gives $\frac{u^2}{\sin^2 b} - \frac{v^2}{\cos^2 b} = 1$. Hence lines parallel to the imaginary axis of the z -plane map into confocal hyperbolas.



8.8 BILINEAR TRANSFORMATION (MOBIUS TRANSFORMATION)

[JNTU 200S, Nov. 2006 (Set No. 3)]

The transformation $w = \frac{az + b}{cz + d}$ where a, b, c, d are complex constants and $ad - bc \neq 0$ is known as "bilinear transformation".

(This is also known as Mobius or linear fractional transformation. The condition $ad - bc \neq 0$ is to ensure that the transformation is conformal, for $\frac{dw}{dz} = \frac{ad - bc}{(cz + d)^2}$ and if $ad - bc = 0$, every point of the z -plane is a critical point).

1. A bilinear transformation is conformal:

Consider the bilinear transformation $w = \frac{az + b}{cz + d}$

Differentiating with respect to z , we have, $w' = \frac{(cz + d)(a) - (az + b)c}{(cz + d)^2} = \frac{ad - bc}{(cz + d)^2}$

Since $ad - bc \neq 0$ we have $w' \neq 0$ (is nowhere zero).

Hence, the mapping defined above is **conformal**.

If $ad - bc = 0$ then $w' = 0$ for all z and every point of z -plane will be critical.

For a choice of constants a, b, c, d we get special cases of Bilinear transformation as

$$(i) \quad w = z + b \quad (\text{Translation}) \quad (ii) \quad w = az + b \quad (\text{Linear transformation})$$

$$(iii) \quad w = az \quad (\text{Rotation}) \quad (iv) \quad w = \frac{1}{z} \quad (\text{Inversion on a unit circle})$$

2. There is a one-one correspondence between all points in the two planes.

$$\text{Let } w = \frac{az + b}{cz + d} \quad \dots(1) \quad \text{be a conformal mapping.}$$

$$\text{Then } z = \frac{-dw + b}{cw - a} \quad \dots(2) \quad \text{is its inverse mapping.}$$

(2) is again a Bilinear transformation.

From (1), it is clear that to each point in the z -plane except $z = -\frac{d}{c}$ there corresponds a unique point in the w -plane.

3. "Invariant" or "Fixed" points of the Bilinear transformation:

For the mapping $w = f(z)$, the points that are mapped onto themselves are called "fixed points" or "invariant points".

Thus, all points such that $z = \frac{az + b}{cz + d}$ are called the invariant or fixed points of the transformation.

$$z = \frac{az + b}{cz + d} \Rightarrow cz^2 - (a - d)z - b = 0. \quad \dots (3)$$

This is a quadratic in z and the roots of this equation are the fixed points of bilinear transformation. If the roots are equal, the bilinear transformation is said to be **parabolic**.

The identity function $w = z$ has every point as fixed.

$$w = \frac{1}{z} \quad \text{has two fixed points.}$$

Rotation has only one fixed point. Translation has no fixed points.

To find the Bilinear transformations whose fixed points are α and β :

The quadratic in z with α, β as roots is $z^2 - (\alpha + \beta)z + \alpha\beta = 0$.

For any complex constant γ ,

$$z^2 - (\alpha + \beta)z + \gamma z - \gamma z + \alpha\beta = 0$$

$$\text{i.e. } z[z - (\alpha + \beta - \gamma)] = \gamma z - \alpha\beta \quad \text{i.e., } z = \frac{\gamma z - \alpha\beta}{z - (\alpha + \beta - \gamma)} = \frac{\gamma z - \alpha\beta}{z - (\alpha + \beta) + \gamma}$$

Thus the bilinear transformations whose fixed points α and β are given by

$$w = \frac{\gamma z - \alpha\beta}{z - (\alpha + \beta) + \gamma}$$

This gives required bilinear transformations for various values of γ .

$$4. w = \frac{az + b}{cz + d}$$

If $a \neq 0$, $w = \frac{z + \frac{b}{a}}{\left(\frac{c}{a}\right)z + \left(\frac{d}{a}\right)}$. Here $\frac{b}{a}$, $\frac{c}{a}$ and $\frac{d}{a}$ are three independent constants.

If $b \neq 0$, we have $w = \frac{\frac{a}{b}z + 1}{\frac{c}{b}z + \frac{d}{b}}$. Here $\frac{a}{b}$, $\frac{c}{b}$ and $\frac{d}{b}$ are three independent constants.

This argument can be continued for $c \neq 0$ and also for the case $d \neq 0$.

To determine three independent constants; in general, we require 3 independent consistent equations. Thus, to decide a bilinear transformation $w = \frac{az + b}{cz + d}$, we require 3 independent (consistent) relations.

5. Every bilinear transformation maps the totality of circles and straight lines in the z -plane onto the totality of circles and straight lines in the w -plane. [JNTU 2005S, Nov. 2008 (Set No. 2)]

Proof : Let $w = \frac{az + b}{cz + d}$ be a bilinear transformation. This can be written as $w = \frac{a}{c} + \frac{bc - ad}{c^2} \frac{1}{\left(z + \frac{d}{c}\right)}$.

This is a combination of the transformations $w_1 = z + \frac{d}{c}$, $w_2 = \frac{1}{w_1}$, $w_3 = \frac{bc - ad}{c^2} w_2$ and $w = \frac{a}{c} + w_3$.

By these transformations we pass from z -plane to w_1 plane, then to w_2 -plane then to w_3 -plane and from w_3 -plane to w -plane. These transformations are the **standard transformations** $w = z + c$, $w = \frac{1}{z}$ and $w = cz$. We know that these transformations map the totality of circles and straight lines in the z -plane onto the totality of circles and straight lines in the w -plane.

6. A bilinear transformation preserves cross ratio property of four points:

Proof : If t_1, t_2, t_3, t_4 are any four numbers then $\frac{(t_1 - t_2)(t_3 - t_4)}{(t_1 - t_4)(t_3 - t_2)}$ is said to be their cross ratio and is denoted by (t_1, t_2, t_3, t_4) .

Let z_1, z_2, z_3, z_4 be points in the z -plane which are mapped onto the points w_1, w_2, w_3, w_4 respectively under the bilinear transformation $w = \frac{az + b}{cz + d}$.

Then, we have

$$w_1 - w_2 = \left(\frac{az_1 + b}{cz_1 + d} \right) - \left(\frac{az_2 + b}{cz_2 + d} \right) = \frac{ad - bc}{(cz_1 + d)(cz_2 + d)} (z_1 - z_2)$$

Similarly we can write $w_3 - w_4$, $w_1 - w_4$, $w_3 - w_2$.

Similarly defining relations for other points, we get

$$\frac{(w_1 - w_2)(w_3 - w_4)}{(w_1 - w_4)(w_3 - w_2)} = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_3 - z_2)}$$

which proves the result.

7. Three given distinct points z_1, z_2, z_3 can always be mapped onto three prescribed distinct points w_1, w_2, w_3 by one and only one, linear fractional transformation $w = f(z)$. This mapping is given implicitly by the equation

$$\frac{w - w_1}{w - w_3} \cdot \frac{w_2 - w_3}{w_2 - w_1} = \frac{z - z_1}{z - z_3} \cdot \frac{z_2 - z_3}{z_2 - z_1}$$

Proof : To find the unique bilinear transformation, consider w as the image of a general point z under this transformation.

Replacing z_4 with z and w_4 with w in the cross ratio of four points

$$\begin{aligned} \frac{(w_1 - w_2)(w_3 - w_4)}{(w_1 - w_4)(w_3 - w_2)} &= \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_3 - z_2)} \\ \Rightarrow \frac{(w_1 - w_2)(w_3 - w)}{(w_1 - w)(w_3 - w_2)} &= \frac{(z_1 - z_2)(z_3 - z)}{(z_1 - z)(z_3 - z_2)} \Rightarrow \frac{(w - w_1)(w_2 - w_3)}{(w - w_3)(w_2 - w_1)} = \frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)} \end{aligned}$$

Hence the statement.

Note. If one of these points is ∞ then the quotient of the two differences containing this point must be replaced by 1.

SOLVED EXAMPLES

Example 1 : Find the bilinear transformation which maps the points $z = 1, i, -1$ onto the points $w = i, 0, -i$.

Hence, find:

- (a) the image of $|z| < 1$, (b) concentric circles $|z| = r$ ($r > 1$),
(c) the invariant points of this transformation.

Solution : Let the points be $z_1 = 1, z_2 = i, z_3 = -1$ and $z_4 = z$ map onto the points $w_1 = i, w_2 = 0, w_3 = -i$ and $w_4 = w$.

We know that cross-ratio of four points is invariant under bilinear transformation.

$$\text{Thus, } \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_3 - z_2)} = \frac{(w_1 - w_2)(w_3 - w_4)}{(w_1 - w_4)(w_3 - w_2)}$$

Substituting the values, we get

$$\frac{(1-i)(-1-z)}{(1-z)(-1-i)} = \frac{(i-0)(-i-w)}{(i-w)(-i-0)}$$

$$\text{i.e., } \frac{w+i}{w-i} = \frac{(z+1)(1-i)}{(z-1)(1+i)}$$

Doing Componendo and Dividendo, we get

$$\frac{2w}{2i} = \frac{(z+1)(1-i) + (z-1)(1+i)}{(z+1)(1-i) - (z-1)(1+i)}$$

$$\therefore w = \frac{1+iz}{1-iz}$$

This is the required bilinear transformation.

$$(a) \ z \text{ can be written as } z = i \left(\frac{1-w}{1+w} \right)$$

$$\therefore \left| \frac{i(1-w)}{1+w} \right| = |z| < 1$$

$$\Rightarrow |i| |1-w| < |1+w|$$

$$\text{or } |1-u-iv| < |1+u+iv|$$

$$\Rightarrow (1-u)^2 + v^2 < (1+u)^2 + v^2$$

$$\Rightarrow u > 0$$

Thus, the interior of the circle $x^2 + y^2 = 1$ in the z -plane is mapped onto the entire half of the w -plane to the right of the imaginary axes.

$$(b) \ |z| = r \text{ is transformed into } \left| i \frac{1-w}{1+w} \right| = |z| = r$$

$$\text{or } |1-w|^2 = r^2 |1+w|^2$$

$$\text{or } (1-w)(1-\bar{w}) = r^2 (1+w)(1+\bar{w})$$

$$\text{or } 1-w-\bar{w}+w\bar{w} = r^2 + r^2(w+\bar{w}) + r^2 w\bar{w}$$

$$\text{or } (1-r^2)w\bar{w} - (1+r^2)(w+\bar{w}) + 1-r^2 = 0$$

$$\text{or } u^2 + v^2 - \frac{1+r^2}{1-r^2} + 1 = 0$$

$$\text{or } \left(u^2 - \frac{1+r^2}{1-r^2} \right) + v^2 = \frac{(1+r^2)^2}{(1-r^2)^2} - 1 = \frac{4r^2}{(1-r^2)^2}$$

$$\text{or } \left| w - \frac{1+r^2}{1-r^2} \right|^2 = \frac{4r^2}{(1-r^2)^2}$$

$$\text{or } \left| w - \frac{1+r^2}{1-r^2} \right| = \frac{2r}{1-r^2}$$

which is a circle in the w -plane with centre $w = \frac{1+r^2}{1-r^2}$ and radius $\frac{2r}{1-r^2}$.

(c) Put $w = z$ to get invariant points

$$\therefore z = \frac{1+iz}{1-iz} \quad \text{or} \quad iz^2 + (i-1)z + 1 = 0$$

$$z = \frac{(1-i) \pm \sqrt{(i-1)^2 - 4i}}{2i} = -\frac{1}{2} \{1+i \pm \sqrt{6i}\}$$

These are the required invariant points.

Example 2 : Find the linear fractional transformation that maps $-1, i, 1$ onto the mapping $0, i, \infty$.

Solution : The required bilinear transformation is given by

$$\frac{(z_1 - z_2)(z_3 - z)}{(z_1 - z)(z_3 - z_2)} = \frac{(w_1 - w_2)(w_3 - w)}{(w_1 - w)(w_3 - w_2)}$$

Putting values,

$$\frac{(-1-i)(1-z)}{(-1-z)(1-i)} = \frac{(0-i)(\infty-w)}{(0-w)(\infty-i)}$$

We obtain $w = -\frac{z+1}{z-1}$ by taking $\frac{(i-\infty)}{(w-\infty)}$ is 1

Example 3 : Find the bilinear transformation that maps the points $(\infty, i, 0)$ into the points $(0, i, \infty)$. [JNTU 2003 (Set No. 1)]

Solution : Take $z_1 = \infty, z_2 = i, z_3 = 0$ and $w_1 = 0, w_2 = i, w_3 = \infty$

Let $z_4 = z$ and $w_4 = w$ so that z and w are a pair of general points. Substituting these, The required bilinear transformation is

$$\frac{(z_1 - z_2)(z_3 - z)}{(z_1 - z)(z_3 - z_2)} = \frac{(w_1 - w_2)(w_3 - w)}{(w_1 - w)(w_3 - w_2)}$$

$$\frac{(\infty - i)(0 - z)}{(\infty - z)(0 - i)} = \frac{(0 - i)(\infty - w)}{(0 - w)(\infty - i)}$$

$$\frac{\infty - w}{\infty - i} \text{ is taken as 1 and } \frac{\infty - i}{\infty - z} \text{ is taken as 1}$$

$$\therefore \frac{z}{i} = \frac{i}{w} \Rightarrow w = \frac{i^2}{z} = \frac{-1}{z}$$

Note. This question is worked by another method below.

Example 4 : Find the bilinear transformation which transform the points $\infty, i, 0$ in the z -plane into $0, i, \infty$ in the w -plane. [JNTU 2003S, 2005 (Set No. 2)]

Solution : Let $z_1 = \infty, z_2 = i, z_3 = 0$ and $w_1 = 0, w_2 = i, w_3 = \infty$.

The required transformation is given by

$$\frac{(\omega - \omega_1)(\omega_2 - \omega_3)}{(\omega_1 - \omega_2)(\omega_3 - \omega)} = \frac{(z - z_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z)}$$

$$\text{i.e. } \frac{(w - 0)(i - \infty)}{(0 - i)(\infty - w)} = \frac{(z - \infty)(i - 0)}{(\infty - i)(0 - z)}$$

$$\text{i.e. } \left(\frac{-w}{i}\right)\left(\frac{i - \infty}{\infty - w}\right) = \left(\frac{-i}{z}\right)\left(\frac{z - \infty}{\infty - i}\right)$$

$$\text{i.e. } \left(\frac{-w}{i}\right)\lim_{n \rightarrow \infty} \left(\frac{i - n}{n - w}\right) = \left(\frac{-i}{z}\right)\lim_{n \rightarrow \infty} \left(\frac{z - n}{n - i}\right) \left(\text{form } \frac{\infty}{\infty}\right)$$

$$\text{i.e. } \left(\frac{-w}{i}\right)\lim_{n \rightarrow \infty} \left(\frac{0 - 1}{1 - 0}\right) = \left(\frac{-i}{z}\right)\lim_{n \rightarrow \infty} \left(\frac{0 - 1}{1 - 0}\right) \quad (\text{L'Hospitals rule})$$

$$\text{i.e. } \left(\frac{-w}{i}\right)(-1) = \left(\frac{-i}{z}\right)(-1)$$

$$\text{or } \frac{w}{i} = \frac{i}{z}$$

$$\text{or } w = \frac{-1}{z}$$

Example 5 : Find the bilinear transformation which maps the points $\infty, i, 0$ in the z -plane into $-1, -i, 1$ in the ω -plane. [JNTU 2003S (Set No. 2)]

Solution : Let $z_1 = \infty, z_2 = i, z_3 = 0$ and $\omega_1 = -1, \omega_2 = -i, \omega_3 = 1$.

The required transformation is

$$\frac{(w - w_1)(w_2 - w_3)}{(w_1 - w_2)(w_3 - w)} = \frac{(z - z_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z)}$$

$$\text{i.e. } \frac{(w + 1)(-i - 1)}{(-1 + i)(1 - w)} = \frac{(z - \infty)(i - 0)}{(\infty - i)(0 - z)} = \left(\frac{-i}{z}\right)\lim_{n \rightarrow \infty} \left(\frac{z - n}{n - i}\right)$$

$$\text{i.e. } \frac{(-1)(w + 1)(1 + i)}{(-1 + i)(1 - w)} = \left(\frac{-i}{z}\right)\left(\frac{0 - 1}{1 - 0}\right) = \left(\frac{-i}{z}\right)(-1) = \frac{i}{z} \quad (\text{L'Hospitals rule})$$

$$\text{i.e. } \frac{1 + w}{1 - w} = \frac{-i(-1 + i)}{(1 + i)z} = \frac{1}{z}$$

Applying componendo and dividendo,

$$\frac{(1 + w) + (1 - w)}{(1 + w) - (1 - w)} = \frac{1 + z}{1 - z} \quad \text{or} \quad \frac{2}{2w} = \frac{1 + z}{1 - z}$$

$$\text{or } \frac{1}{w} = \frac{1 + z}{1 - z} \quad \text{or} \quad w = \frac{1 - z}{1 + z}$$

Example 6 : Determine the linear fractional transformation that maps $z_1 = 0$, $z_2 = 1$, $z_3 = \infty$ onto $w_1 = -1$, $w_2 = -i$, $w_3 = 1$ respectively.

Solution : The required bilinear transformation is

$$\frac{(w_1 - w_2)(w_3 - w)}{(w_1 - w)(w_3 - w_2)} = \frac{(z_1 - z_2)(z_3 - z)}{(z_1 - z)(z_3 - z_2)}$$

Substituting the values, we get $\frac{(-1+i)(1-w)}{(-1-w)(1+i)} = \frac{(-1)(\infty-z)}{(-z)(\infty-1)}$

$$\Rightarrow w = \frac{z-i}{z+i} \text{ by replacing } \frac{(1-\infty)}{(z-\infty)} \text{ by } 1$$

This is sometimes known as **Cayley transformation**.

Example 7 : Find the bilinear transformation which maps the points $(-1, 0, 1)$ into the points $(0, i, 3i)$ [JNTU A 2005, 2007S, Aug. 2007S, 2008S, JNTU (H) Nov. 2009 (Set No. 4)]

Solution : Given $z_1 = -1$, $z_2 = 0$, $z_3 = 1$, $w_1 = 0$, $w_2 = i$, $w_3 = 3i$

We know that $\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$

$$\Rightarrow \frac{(w-0)(i-3i)}{(0-i)(3i-w)} = \frac{(z+1)(0-1)}{(-1-0)(1-z)}$$

$$\Rightarrow \frac{2w}{3i-w} = \frac{z+1}{1-z}$$

$$\Rightarrow (2w)(1-z) = (z+1)(3i-w)$$

$$\Rightarrow (2w)(1-z) = 3iz + 3i - wz - w$$

$$\Rightarrow w(2-2z+z+1) = 3i(z+1)$$

$$\Rightarrow w(-z+3) = 3i(z+1) \Rightarrow w = \frac{3i(z+1)}{(3-z)}$$

Example 8 : Find the linear fractional transformation that maps the points $z_1 = -2$, $z_2 = 0$, $z_3 = 2$ onto the points $w_1 = \infty$, $w_2 = 1/4$, $w_3 = 3/8$ respectively.

Solution : The required bilinear transformation is

$$\frac{(z_1 - z_2)(z_3 - z)}{(z_1 - z)(z_3 - z_2)} = \frac{(w_1 - w_2)(w_3 - w)}{(w_1 - w)(w_3 - w_2)}$$

Substituting the values,

$$\frac{(-2-0)(2-z)}{(-2-z)(2-0)} = \frac{(\infty-1/4)(3/8-w)}{(\infty-w)(3/8-1/4)}$$

Simplifying $w = \frac{z+1}{2z+4}$

Example 9 : Find the bilinear transformation that maps the points $0, 1, i$ in z -plane onto the points $1 + i, -i, 2 - i$ in the w -plane.

Solution : The required bilinear transformation is

$$\frac{(w_1 - w_2)(w_3 - w)}{(w_1 - w)(w_3 - w_2)} = \frac{(z_1 - z_2)(z_3 - z)}{(z_1 - z)(z_3 - z_2)}$$

Putting values, we get $\frac{(1+i-i)(2-i-w)}{(1+i-w)(2-i+i)} = \frac{(0-1)(i-z)}{(0-z)(i-1)}$

$$\Rightarrow \frac{(1+2i)(2-i-w)}{2(1+i-w)} = (i-1) \left(\frac{i-z}{z} \right)$$

$$\Rightarrow \frac{(2-i-w)}{(1+i-w)} = \frac{2(3i+1)}{5} \left(\frac{i-z}{z} \right)$$

$$\Rightarrow 5z(2-i-w) = 2(3i+1)(1+i-w)(i-z)$$

$$\Rightarrow w = \frac{(6i+2)(1+i)(i-z) - (2-i)5z}{-5z + (6i+2)(i-z)}$$

$$\therefore w = \frac{z(6+3i) + (8+4i)}{z(7+6i) + (6-2i)}$$

Example 10 : Find the linear fractional (bilinear) transformation that maps $z_1 = -1, z_2 = 0, z_3 = 1$ onto $w_1 = -1, w_2 = -i, w_3 = 1$ respectively.

Solution : The required bilinear transformation is given by

$$\frac{(z_1 - z_2)(z_3 - z)}{(z_1 - z)(z_3 - z_2)} = \frac{(w_1 - w_2)(w_3 - w)}{(w_1 - w)(w_3 - w_2)}$$

Putting values, $\frac{(-1-0)(1-z)}{(-1-z)(1-0)} = \frac{(-1+i)(1-w)}{(-1-w)(1+i)}$

Simplifying $w = \frac{z-i}{-iz+1}$

Example 11 : Find the bilinear transformation that maps the points $(0, 1, \infty)$ in z -plane onto the points $(-1, -2, -i)$ in the w -plane. [JNTU April, Nov. 2006 (Set No. 2)]

Solution : Take $z_1 = 0, z_2 = 1, z_3 = \infty$ and $w_1 = -1, w_2 = -2, w_3 = -i$

The required bilinear transformation is

$$\frac{(w_1 - w_2)(w_3 - w)}{(w_1 - w)(w_3 - w_2)} = \frac{(z_1 - z_2)(z_3 - z)}{(z_1 - z)(z_3 - z_2)}$$

Putting values,

$$\frac{(-1+2)(-i-w)}{(-1-w)(-i+2)} = \frac{(0-1)(z_3-z)}{(0-z)(z_3-1)}$$

$$\text{i.e.} \quad \frac{-(1+w)}{-(1+w)(2-i)} = \frac{1}{z} \left[\because \frac{z_3 - z}{z_3 - 1} \text{ is taken as } 1 \right]$$

Solving for w , we get

$$z(1+w) = (1+w)(2-i)$$

or

$$w = \frac{iz + (1-2)}{-z + (2-i)}$$

Which is the required transformation.

Example 12 : Find the bilinear transformation that maps the points $1, i, -1$ into the points $2, i, -2$. [JNTU 2004 (Set No. 2)]

Solution : Let $w = \frac{az+b}{cz+d}$... (1) be the required bilinear transformation.

Substituting the corresponding values of z and w in (1), we get

$$2 = \frac{a+b}{c+d}, \quad i = \frac{ai+b}{ci+d}, \quad -2 = \frac{-a+b}{-c+d}$$

$$\text{Hence} \quad a+b = 2c+2d \quad \dots (2)$$

$$ai+b = -c+id \quad \dots (3)$$

$$-a+b = 2c-2d \quad \dots (4)$$

$$(2) + (4) \text{ gives } 2b = 4c \text{ i.e., } b = 2c$$

$$(2) - (4) \text{ gives } 2a = 4d \text{ i.e., } a = 2d$$

$$\text{Substituting in (3), } 2id + 2c = -c + id$$

$$\Rightarrow d = -\frac{3c}{i} \text{ and } a = -\frac{6c}{i}$$

Substituting the values of a, b, d in (1), we obtain

$$w = \frac{-\frac{6c}{i}z + 2c}{cz - \frac{3c}{i}} = \frac{-6cz + 2ic}{icz - 3c} = \frac{-6z + 2i}{iz - 3} \text{ which is the required bilinear transformation.}$$

Example 13 : Find the fixed points of the transformation

$$(i) w = \frac{2i-6z}{iz-3} \quad (ii) w = \frac{6z-9}{z} \quad (iii) w = \frac{z-1}{z+1} \quad [\text{JNTU (A) Dec 2009 (Set No. 4)}]$$

Solution : The fixed points of the transformation are such that the image of z is z itself.

$$(i) \text{ Putting } w = z \text{ in the given transformation, we get } z = \frac{2i-6z}{iz-3}$$

$$\text{i.e.} \quad iz^2 - 3z = 2i - 6z$$

$$\text{i.e.,} \quad iz^2 + 3z - 2i = 0$$

$$\therefore z = \frac{-3 \pm \sqrt{9 - 4(i)(-2i)}}{2i} = \frac{-3 \pm \sqrt{9-8}}{2i} = \frac{-3 \pm 1}{2i}$$

$$\text{i.e. } z = \frac{-3+1}{2i} = \frac{-1}{i} = i \quad \text{and} \quad z = \frac{-3-1}{2i} = \frac{-2}{i} = 2i$$

Hence fixed points are $i, 2i$.

(ii) The fixed points are given by $w = z$

$$\text{i.e. } z = \frac{6z-9}{z}$$

$$\text{i.e. } z^2 - 6z + 9 = 0 \quad \text{i.e. } (z-3)^2 = 0$$

$$\text{or } z = 3, 3$$

Hence fixed points are $3, 3$

Note. The above transformation is *parabolic* since it contains only one fixed point.

(iii) The fixed points are given by $w = z$.

$$\text{i.e. } z = \frac{z-1}{z+1}$$

$$\text{i.e. } z^2 + z - z + 1 = 0$$

$$\text{i.e. } z^2 - i^2 = 0 \quad \text{or } (z-i)(z+i) = 0$$

$$\therefore z = \pm i$$

Hence fixed points are $i, -i$.

Example 14 : Determine the bilinear transformation that maps the points $(1-2i, 2+i, 2+3i)$ into the points $(2+i, 1+3i, 4)$ [JNTU 2003, 2004 (Set No. 4), 2004S (Set No. 2)]

Solution : Take $z_1 = 1-2i, z_2 = 2+i, z_3 = 2+3i$

and $w_1 = 2+i, w_2 = 1+3i, w_3 = 4$

Let $z_4 = z$ and $w_4 = w$ so that z and w are a pair of general points. Substituting these in

$$\frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_3 - z_2)} = \frac{(w_1 - w_2)(w_3 - w_4)}{(w_1 - w_4)(w_3 - w_2)}$$

$$\text{we get } \frac{(1-2i-2-i)(2+3i-z)}{(1-2i-z)(2+3i-2-i)} = \frac{(2+i-1-3i)(4-w)}{(2+i-w)(4-i-3i)}$$

$$\text{i.e. } \frac{2+3i-z}{1-2i-z} = -\frac{1}{3} \left(\frac{4-w}{2+i-w} \right) = \frac{w-4}{3(2+i-w)}$$

$$\text{Solving, } w = \frac{z(10+3i)-7-6i}{4z-7-7i}$$

Example 15 : Find the bilinear transformation which maps the points $(-i, 0, i)$ into the points $(-1, i, 1)$ respectively. [JNTU 2005S, Nov. 2006, Nov. 2008S (Set No. 3)]

Solution : Let $z_1 = -i, z_2 = 0, z_3 = i$ and $w_1 = -1, w_2 = i, w_3 = 1$

The required transformation is given by

$$\frac{(w_1 - w_2)(w_3 - w)}{(w_1 - w)(w_3 - w_2)} = \frac{(z_1 - z_2)(z_3 - z)}{(z_1 - z)(z_3 - z_2)}$$

Putting values,

$$\frac{(-1-i)(1-w)}{(-1-w)(1-i)} = \frac{(-i-0)(i-z)}{(-i-z)(i-0)}$$

$$\text{i.e. } \frac{(1+i)(1-w)}{(1-i)(i+w)} = \frac{i(i-z)}{i(i+z)} = \frac{i-z}{i+z}$$

$$\text{i.e. } \frac{1-w}{1+w} = \frac{i-z}{i+z} \cdot \frac{1-i}{1+i} = \frac{i-z}{i+z} (-i) = \frac{zi+1}{z+i}$$

Doing componendo and dividendo, we get

$$\frac{(1-w)+(1+w)}{(1-w)-(1+w)} = \frac{(zi+1)+(z+i)}{(zi+1)-(z+i)} = \frac{(1+i)(z+1)}{(1-i)(-z+1)}$$

$$\text{i.e. } \frac{2}{-2w} = \frac{i(z+1)}{(-z+1)}$$

$$\text{or } -w = \frac{-z+1}{i(z+1)} = \frac{i(-z+1)}{i^2(z+1)}$$

$$\text{or } w = \frac{-iz+i}{z+1} \text{ is the required transformation.}$$

Example 16 : Find the bilinear transformation which maps vertices $(1+i, -i, 2-i)$ of the triangle T of the z -plane into the points $(0, 1, i)$ of the w -plane. [JNTU 2005S (Set No. 2)]

Solution : The required transformation is

$$\frac{(w_1 - w_2)(w_3 - w)}{(w_1 - w)(w_3 - w_2)} = \frac{(z_1 - z_2)(z_3 - z)}{(z_1 - z)(z_3 - z_2)}$$

Putting values,

$$\frac{(0-1)(i-w)}{(0-w)(i-1)} = \frac{(1+i-i)(2-i-z)}{(1+i-z)(2-i+i)}$$

$$\text{or } \frac{i-w}{w(i-1)} = \frac{(1+2i)(2-i-z)}{2(1+i-z)} \quad \text{or } \frac{w-i}{w} = \frac{(1+2i)(2-i-z)}{2(1+i-z)} (1-i)$$

Solving for w ,

$$1 - \frac{i}{w} = \frac{(1+2i)(1-i)(2-i-z)}{2(1+i-z)} = \frac{7-i-3z-iz}{2(1+i-z)}$$

$$\text{or } \frac{i}{w} = 1 - \frac{7-i-3z-iz}{2(1+i-z)} = \frac{-5+3i+z(1+i)}{2(1+i-z)}$$

$$\text{or } w = \frac{2i(1+i-z)}{z(1+i)+(-5+3i)} = \frac{z(-2i)+2(i-1)}{z(1+i)+(-5+3i)}$$

Example 17 : Find the bilinear transformation which transforms the points $z = 2, 1, 0$ into $w = 1, 0, i$ respectively. [JNTU 2006 (Set No. 3), 2008S (Set No. 4)]

Solution : Let the transformation be $w = \frac{az+b}{cz+d}$... (1)

Substituting the three pairs of values for z and w in (1), we get

$$1 = \frac{2a+b}{2c+d} \Rightarrow 2a+b = 2c+d \quad \dots (2)$$

$$0 = \frac{a+b}{c+d} \Rightarrow a+b = 0 \quad \text{or} \quad a = -b \quad \dots (3)$$

$$i = \frac{b}{d} \Rightarrow b = id \quad \text{or} \quad d = \frac{b}{i} = -bi \quad \dots (4)$$

Substituting (3) and (4) in (2), we get

$$\begin{aligned} -2b+b &= 2c-bi \Rightarrow -b+bi = 2c \\ \Rightarrow b(i-1) &= 2c \quad \text{or} \quad c = \frac{b(i-1)}{2} \quad \dots (5) \end{aligned}$$

Substituting (3), (4) and (5) in (1), we get

$$w = \frac{-bz+b}{\frac{b}{2}(i-1)z-bi} = \frac{-z+1}{\left(\frac{i-1}{2}\right)z-i} = \frac{-2z+2}{(i-1)z-2i}$$

Example 18 : Find the bilinear transformation which maps the points $(2, i, -2)$ into the points $(1, i, -1)$. [JNTU 2007S (Set No. 1)]

Solution : Let the points $z_1=2, z_2=i, z_3=-2$ and $z_4=z$ map onto the points $w_1=1, w_2=i, w_3=-1$ and $w_4=w$.

The required transformation is given by $\frac{(z_1-z_2)(z_3-z)}{(z_1-z)(z_3-z_2)} = \frac{(w_1-w_2)(w_3-w)}{(w_1-w)(w_3-w_2)}$

Substituting the values, we get

$$\frac{(2-i)(-2-z)}{(2-z)(-2-i)} = \frac{(1-i)(-1-w)}{(1-w)(-1-i)}$$

$$\text{i.e.} \quad \frac{(2-i)(2+z)}{(2-z)(2+i)} = \frac{(1-i)(1+w)}{(1-w)(1+i)}$$

$$\text{or} \quad \frac{1+w}{1-w} = \left(\frac{2+z}{2-z}\right)\left(\frac{2-i}{2+i}\right)\left(\frac{1+i}{1-i}\right)$$

$$\begin{aligned} \text{or} \quad \frac{w-1}{w+1} &= \left(\frac{z-2}{z+2}\right)\left(\frac{2+i}{2-i}\right)\left(\frac{1-i}{1+i}\right) = \left(\frac{z-2}{z+2}\right)\left(\frac{4-1+4i}{4+1}\right)\left(\frac{1-1-2i}{1+1}\right) \\ &= \frac{(4-3i)(z-2)}{5(z+2)} \end{aligned}$$

By Componendo and Dividendo, we get

$$\frac{(w-1) + (w+1)}{(w-1) - (w+1)} = \frac{(4-3i)(z-2) + 5(z+2)}{(4-3i)(z-2) - 5(z+2)}$$

$$\text{i.e. } \frac{2w}{-2} = \frac{3z(3-i) + 2(1+3i)}{-iz(3-i) + 6(i-3)} = \frac{3z(3-i) + 2i(3-i)}{-iz(3-i) - 6(3-i)}$$

$$\text{or } -w = \frac{(3z+2i)(3-i)}{-(iz+6)(3-i)} \quad \text{or } w = \frac{3z+2i}{iz+6}$$

Example 19 : Determine the Bilinear transformation which maps $z = 0, -i, 2i$ into $w = 5i, \infty, -i/3$. [JNTU Nov. 2008 (Set No.3)]

Solution : Let $w = \frac{az+b}{cz+d} \dots (1)$

be the required bilinear transformation.

Substituting the corresponding values of z and w in (1), we get

$$5i = \frac{b}{d} \Rightarrow b = 5id \dots (2)$$

$$-ic + d = 0 \Rightarrow c = \frac{d}{i} \dots (3)$$

$$\begin{aligned} \text{and } -\frac{i}{3} &= \frac{2ia+b}{2ic+d} \\ \Rightarrow 2c - id &= 6ia + 3b \dots (4) \end{aligned}$$

Substituting (2) and (3) in (4), we get

$$\frac{2d}{i} - id = 6ia + 15id \Rightarrow 3d = -6a - 15d \Rightarrow a = -3d$$

Substituting the values of a, b, c in (1), we get

$$w = \frac{(-3d)z + 5id}{(d/i)z + d} = \frac{-3z + 5i}{(z/i) + 1} = \frac{-(3iz + 5)}{z + i}$$

which is the required bilinear transformation.

An Alternative method :

Taking $z_1 = 0, z_2 = -i, z_3 = 2i$

$w_1 = 5i, w_2 = \infty, w_3 = -i/3$ and substituting in the formula,

$$\frac{(w_1 - w_2)(w_3 - w)}{(w_1 - w)(w_3 - w_2)} = \frac{(z_1 - z_2)(z_3 - z)}{(z_1 - z)(z_3 - z_2)}$$

$$\text{we get, } \frac{(5i - \infty)(-\frac{i}{3} - w)}{(5i - w)(-\frac{i}{3} - \infty)} = \frac{(0 + i)(2i - z)}{(0 - z)(2i + i)} = \frac{(i)(2i - z)}{(-z)(3i)}$$

$$\lim_{n \rightarrow \infty} \frac{\left(-\frac{i}{3} - w\right)}{(5i - w)} = \lim_{n \rightarrow \infty} \frac{5i - n}{-\frac{i}{3} - n} = \frac{2i - z}{-3z} \quad (\text{using L' Hospital's rule})$$

$$zi + 3zw = 10i^2 - 5iz - 2iw + wz$$

$$w(2z + 2i) = -10 - 6iz$$

$$W = \frac{-5 - 3iz}{z + i}$$

Example 20 : Find the bilinear transformations whose fixed points are $i, -i$.

Solution : We know that the Bilinear transformations having α and β as fixed points are

$$\text{given by } w = \frac{\gamma z - \alpha\beta}{z - (\alpha + \beta) + \gamma} \quad \text{for various values of } \gamma.$$

For $\alpha = i, \beta = -i$, we have

$$w = \frac{\gamma z - 1}{z + \gamma}$$

$$\text{For } \gamma = 0, w = -\frac{1}{z}; \gamma = 1, w = \frac{z-1}{z+1}, \gamma = 2, w = \frac{2z-1}{z+2}$$

Example 21 : Find the bilinear transformation whose fixed points are $1, i$ and maps 0 to -1 .

Solution : The bilinear transformation with α, β as fixed points is given by

$$w = \frac{\gamma z - \alpha\beta}{z - (\alpha + \beta) + \gamma}$$

for various values of γ .

For $\alpha = 1, \beta = i$, we have

$$w = \frac{\gamma z - i}{z - 1 - i + \gamma}$$

Since $z = 0$ is mapped to $w = -1$, we have

$$-1 = \frac{0 - i}{0 - 1 - i + \gamma} \Rightarrow \gamma = 2i + 1$$

Thus the required bilinear transformation is

$$w = \frac{(2i+1)z - i}{z - 1 - i + (2i+1)} = \frac{(2i+1)z - i}{z + i}$$

EXERCISE

1. Under the transformation $w = \frac{1}{z}$, find the image of the infinite strip $\frac{1}{8} < y < \frac{1}{4}$.
2. Find the image of the circle $|z| = 2$ under the transformation $w = z + 3 + 2i$.
3. Find the map of the circle $|z| = c$ under the transformation $w = z - 2 + 4i$.
4. (i) Find the image of $|z| = 2$ under the transformation $w = 3z$.
(ii) Find the points at which $w = \cosh z$ is not conformal. [JNTU 2003S(Set No. 1)]
5. Prove that under the transformation $w = \frac{1}{z}$, the image of the lines $y = x - 1$ and $y = 0$ are the circle $u^2 + v^2 - u - v = 0$ and the line $v = 0$ respectively. [JNTU 2004 (Set No. 1)]
6. Determine the region into which the rectangular region in the z -plane bounded by the lines $x = 0$, $y = 0$, $x = 2$ and $y = 1$ is mapped under the transformation $w = z + (1 - 2i)$.
7. Find the angle of rotation at the point $z = z + i$ when the transformation is $w = z^2$. Also find the scale factor of the transformation at that point.
8. Find the bilinear transformation which transform the points $-2, 0, 2$ in the z -plane into $0, i, -i$ respectively in the w -plane.
9. Find the bilinear transformation which maps the points $0, 1, \infty$ in the z -plane into $-5, -1, 3$ respectively in the w -plane. What are the invariant points in this transformation?
10. Determine the bilinear transformation that maps the points $1 - 2i, 2 + i, 2 + 3i$ respectively into $2 + 2i, 1 + 3i, 4$. [JNTU 2001S]

ANSWERS

1. Region common to the circles $u^2 + (v + 4)^2 = 16$, $u^2 + (v + 2)^2 = 4$
2. Circle with centre $(3, 2)$ and radius 2
3. $(u + 2)^2 + (v - 4)^2 = c^2$
6. The rectangle bounded by $u = 1$, $u = 3$, $v = -1$ and $v = -2$
7. $\tan^{-1} \frac{1}{2}, 2\sqrt{5}$
8. $w = \frac{i(z+2)}{2-3z}$
9. $w = \frac{3z-5}{z+1}; z = 1 \pm i2$